



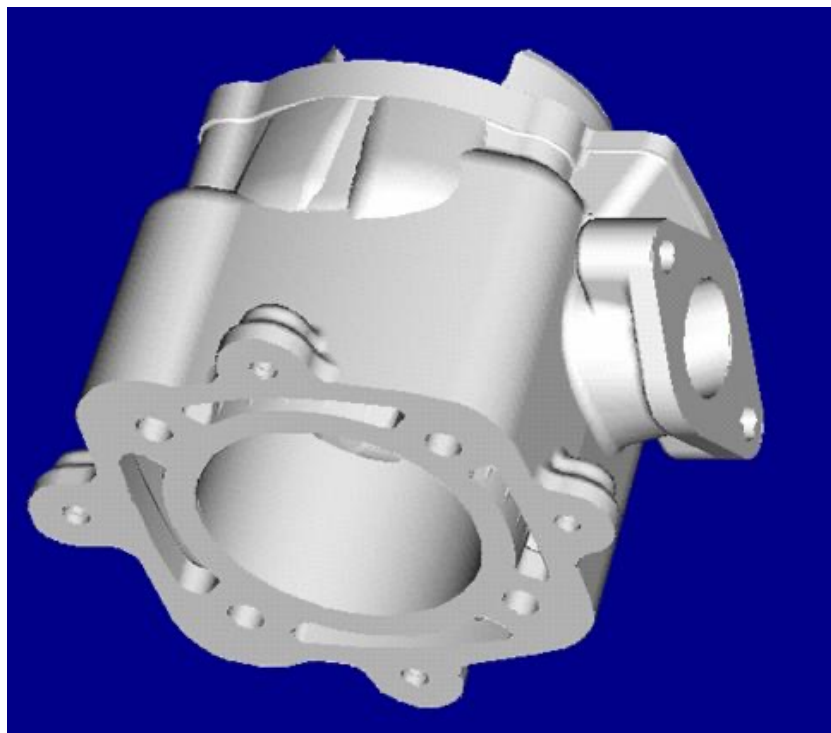
Introduction to Geometric Modeling

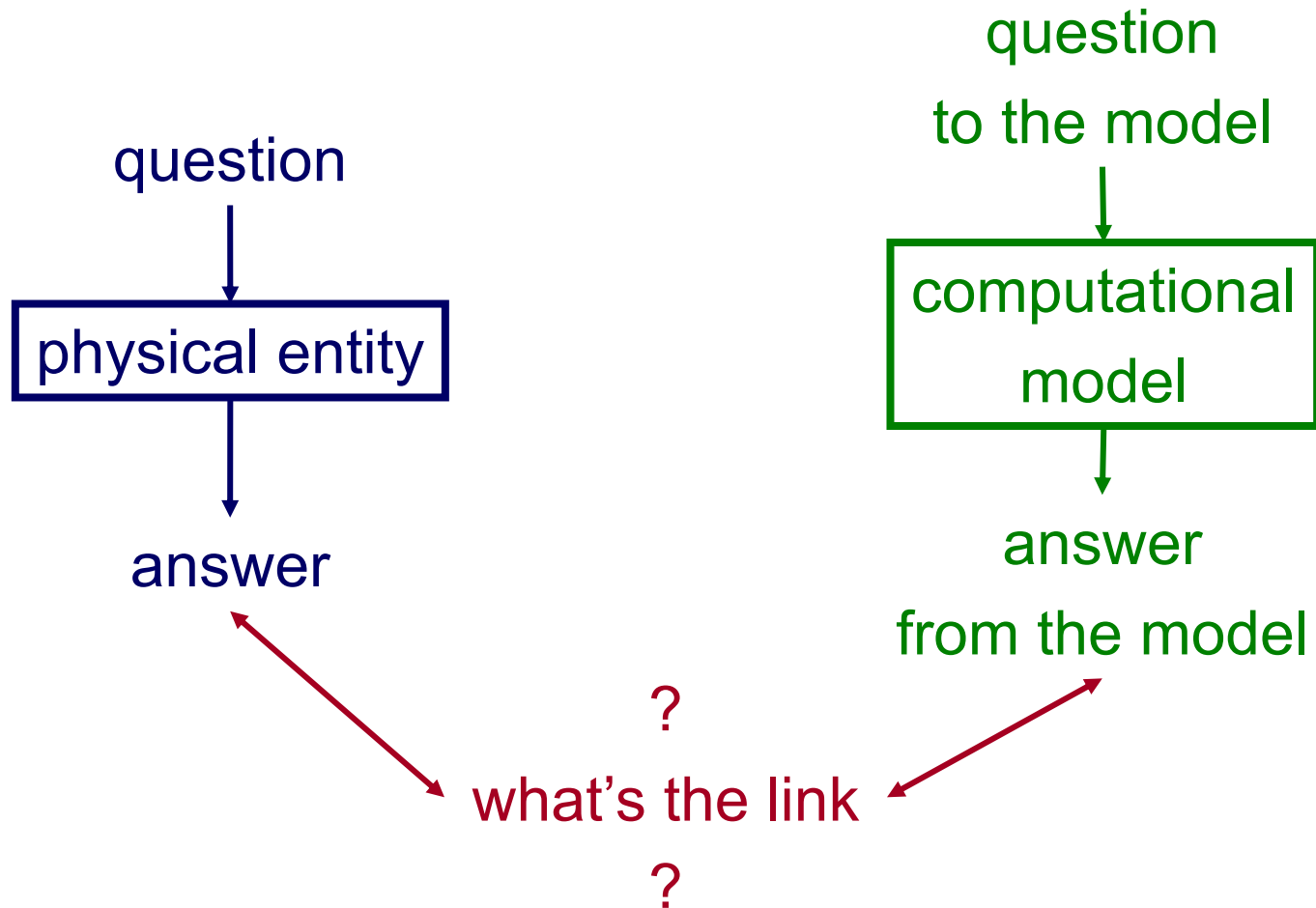
Agoston M.K. (2005)
Computer Graphics and Geometric Modeling
Chapters: 5, 11 and 12

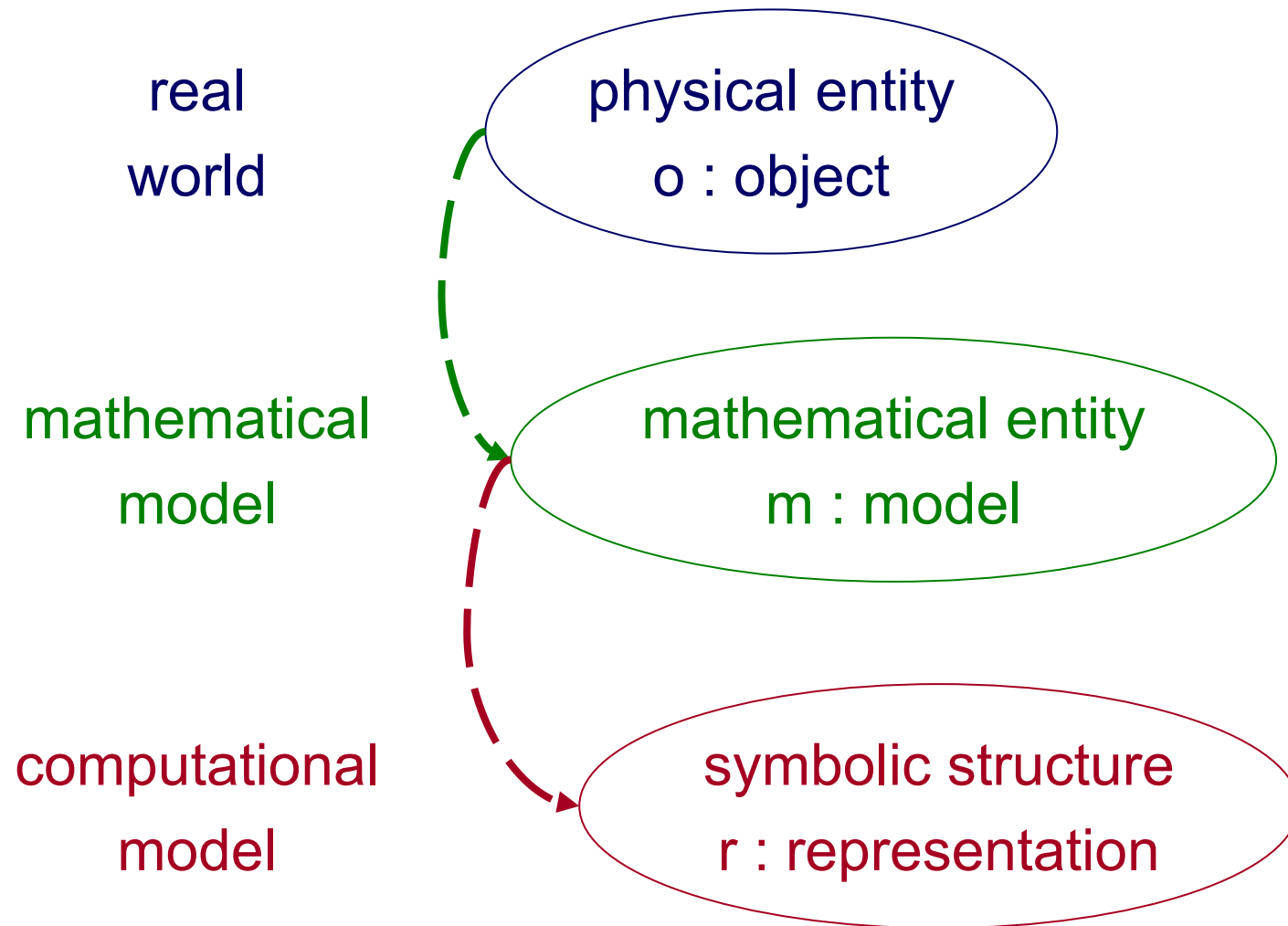


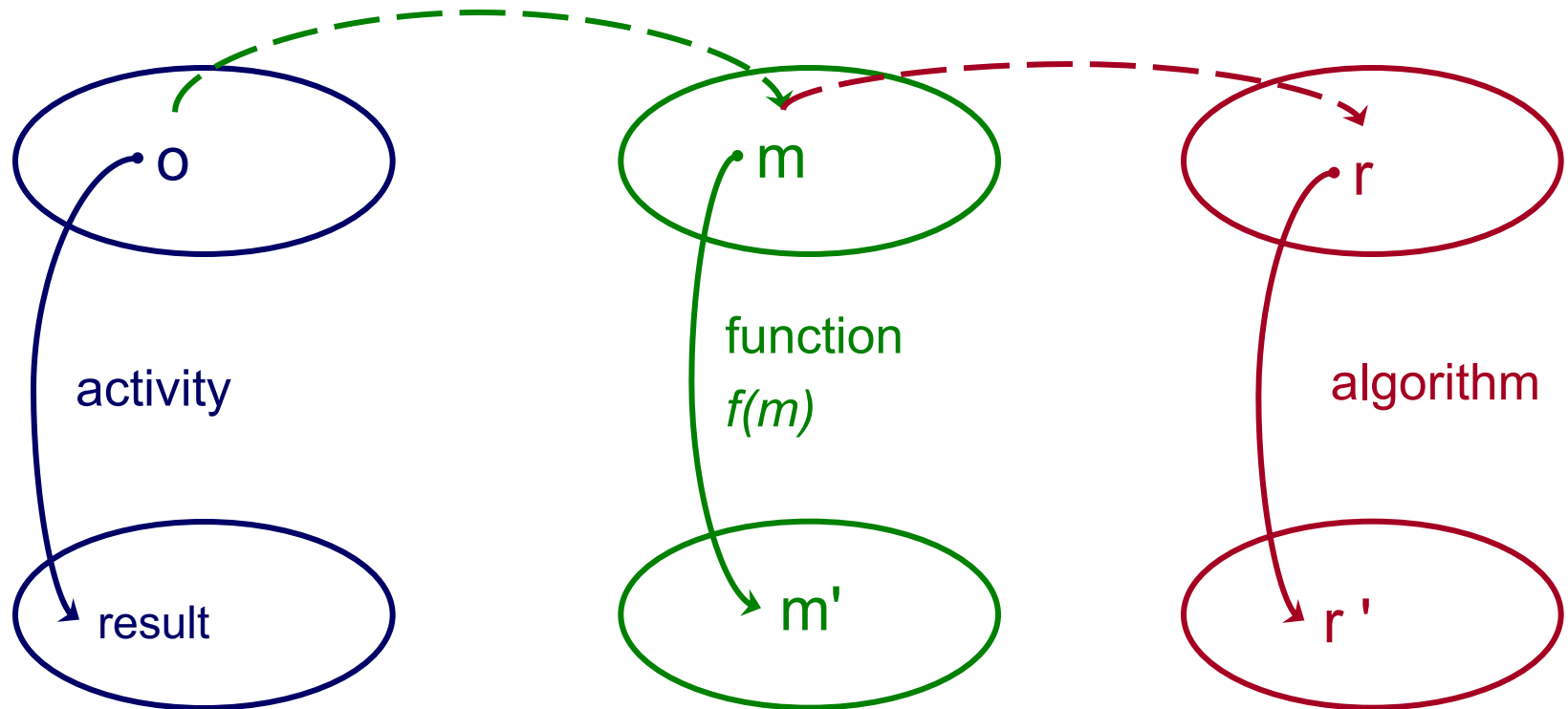
Three possible geometric elements:

- Solid
- Surface
- Line











COMPUTATIONAL MODEL

A physical object (a solid) is mathematically defined as a point-set in the three dimensional Euclidean space (R^3). This point-set must guarantee the following properties:

- ***Rigidity***: Invariant with respect to location/orientation.
- ***Homogeneous 3-dimensionality***: The boundaries must be in contact with the interior.
- ***Boundary determinism***: The boundary must contain the solid and hence determine the solid distinctly.
- ***Finiteness and finite describability***: Size is finite and a finite amount of information can describe the solid.
- ***Closure under rigid motion and Boolean operations***: Movement and Boolean operations should produce other valid solids.



COMPUTATIONAL MODEL

A computational model is characterized by a symbolic structure, called “**representation scheme**”.

A representation scheme may have the following properties:

Formal Properties

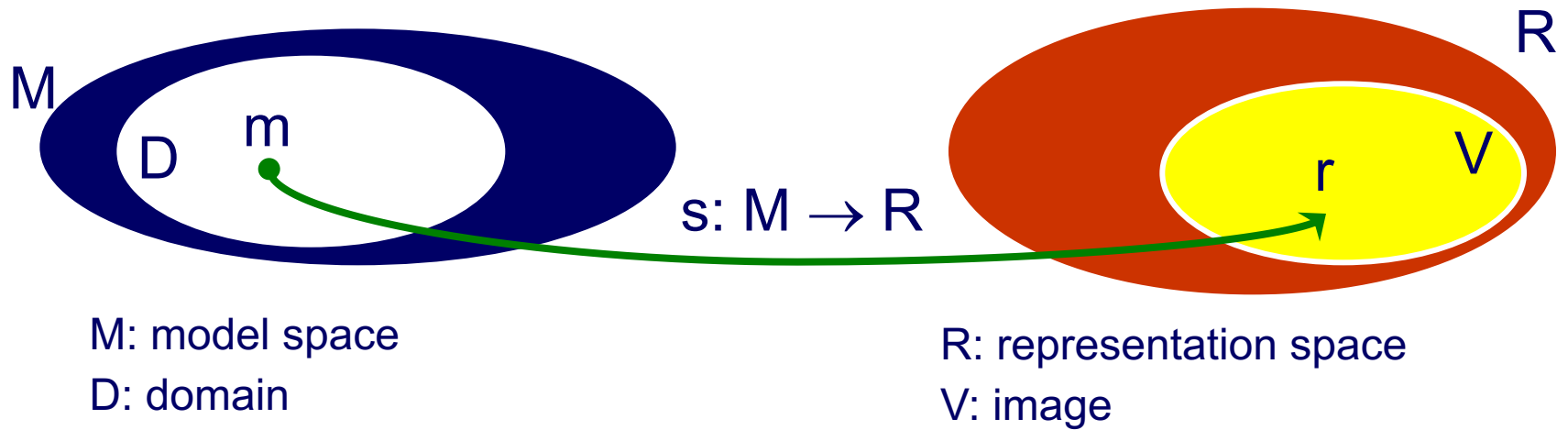
- domain;
- validity;
- completeness;
- uniqueness.

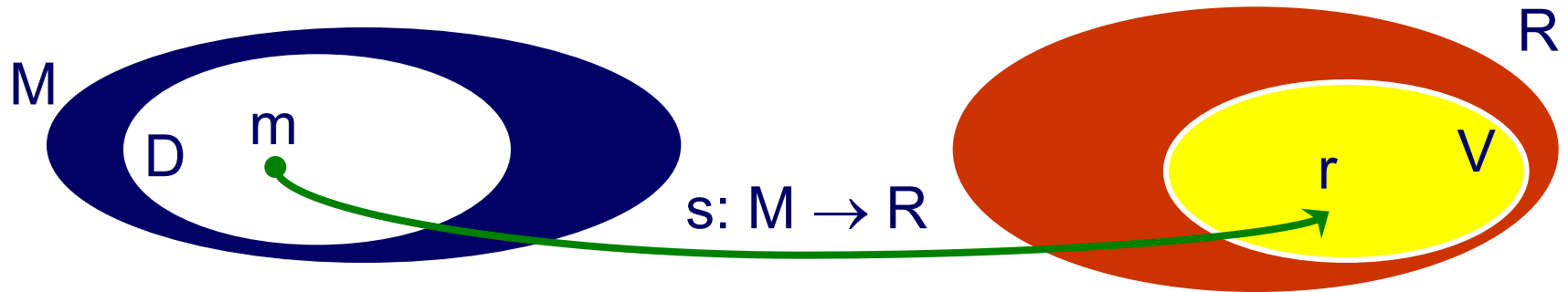
Desirable Properties

- conciseness;
- ease of creation;
- efficiency in application.



Domain



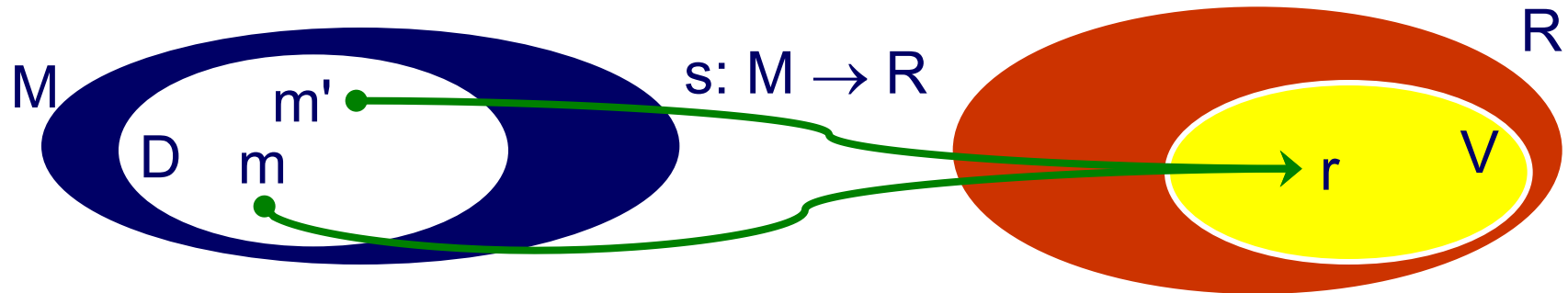


Syntactic correctness: $\forall m \in D \subseteq M \Rightarrow \exists r \in V \subseteq R$

Correctness of semantics: $\forall r \in V \subseteq R \quad \exists m \in D \subseteq M : r = s(m)$

Validity

A representation scheme is valid if and only if it is syntactically and semantically correct.



Ambiguity: a representation scheme is ambiguous if
$$\exists m, m' \in D : r = s(m) = s(m') \in V$$

Completeness: a representation scheme is complete if all its representations are valid and unambiguous

Uniqueness: a representation scheme is unique if
$$\forall m \in D \quad \exists 1! r \in V : r = s(m)$$

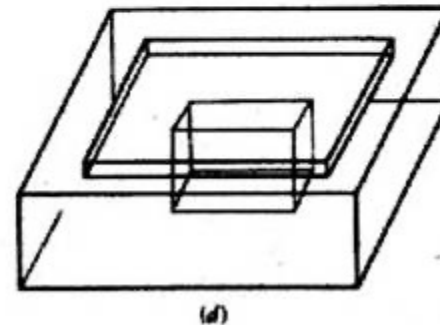
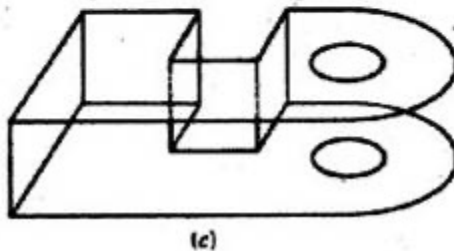
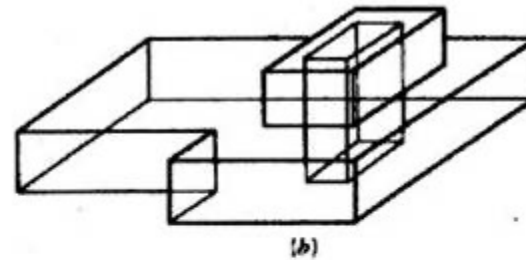
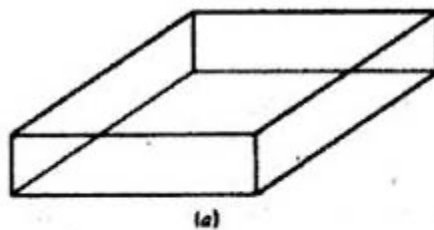


In order to represent solids (rigid bodies), the most used representation schemes are:

- wire frame
- parameterized primitive instancing;
- spatial occupancy enumeration;
- cell decomposition;
- sweeping;
- constructive solid geometry;
- boundary representation.



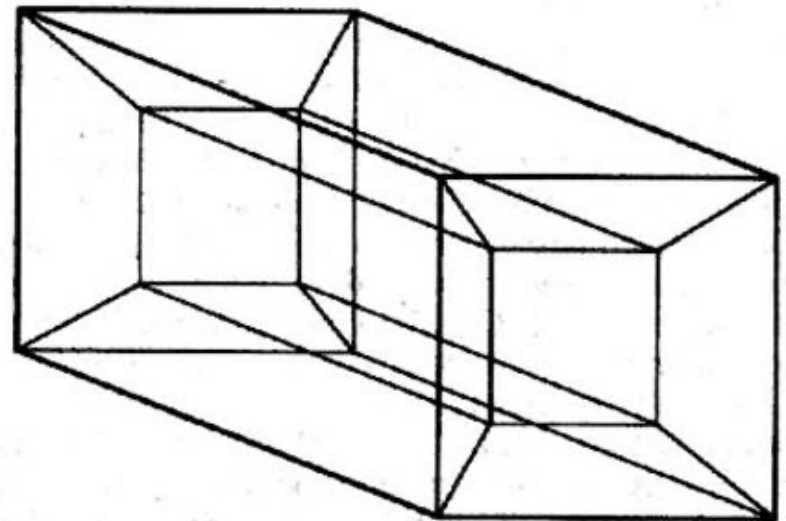
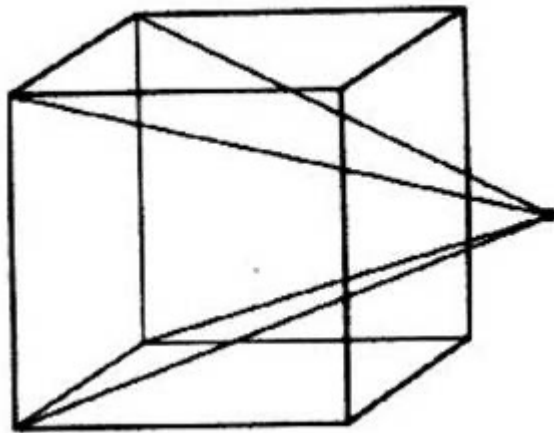
A *wireframe* representation of an object is done using edges and vertices.





Wireframe disadvantages:

- Nonsense object
- Ambiguity
- Lengthy and verbose to define

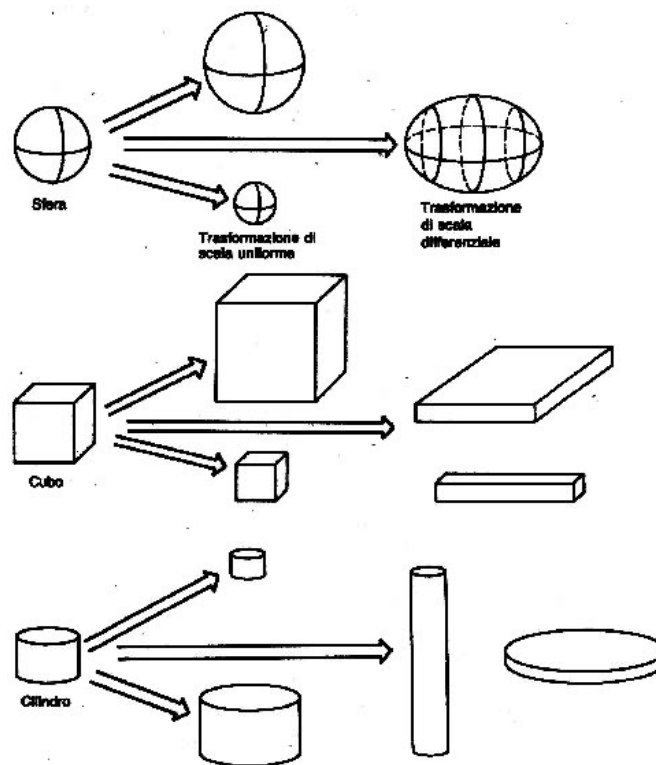




SOLID REPRESENTATION SCHEME

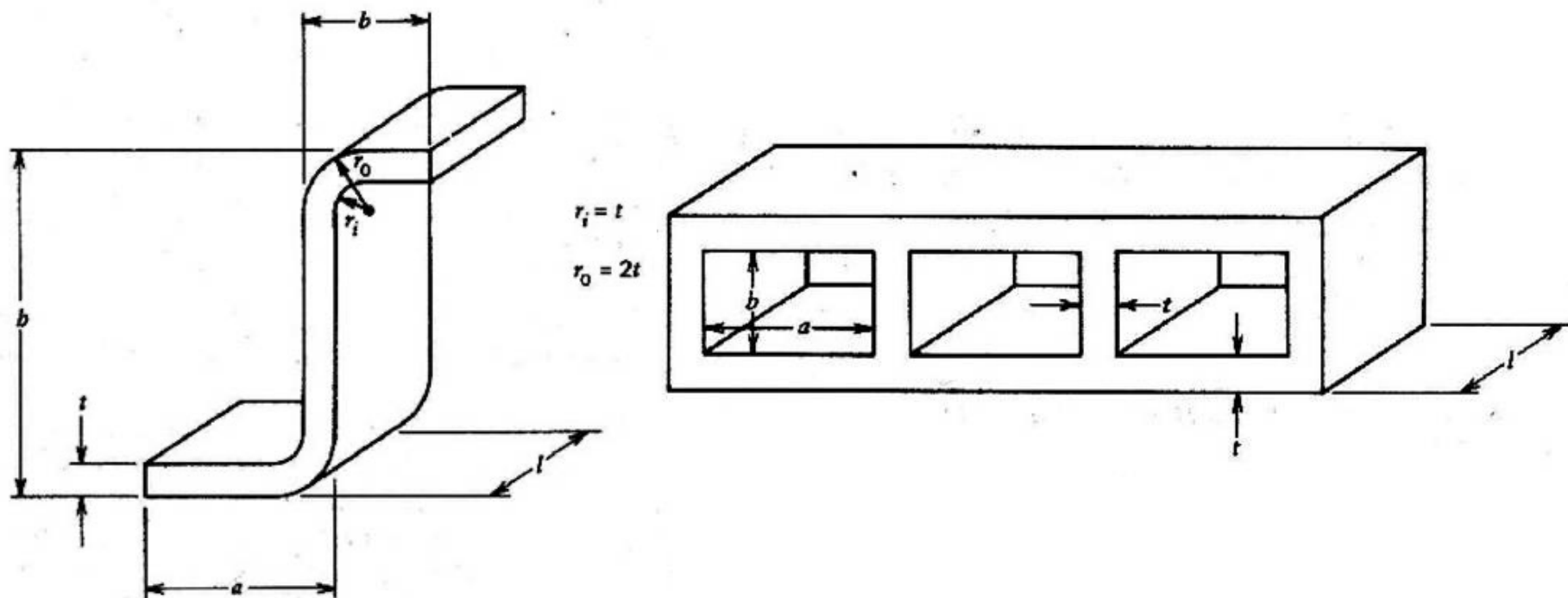
Parameterized primitive instantiation

This scheme is based on notion of families of object, each member of a family distinguishable from the other by a few parameters. Each object family is called a *generic primitive*, and individual objects within a family are called *primitive instances*.



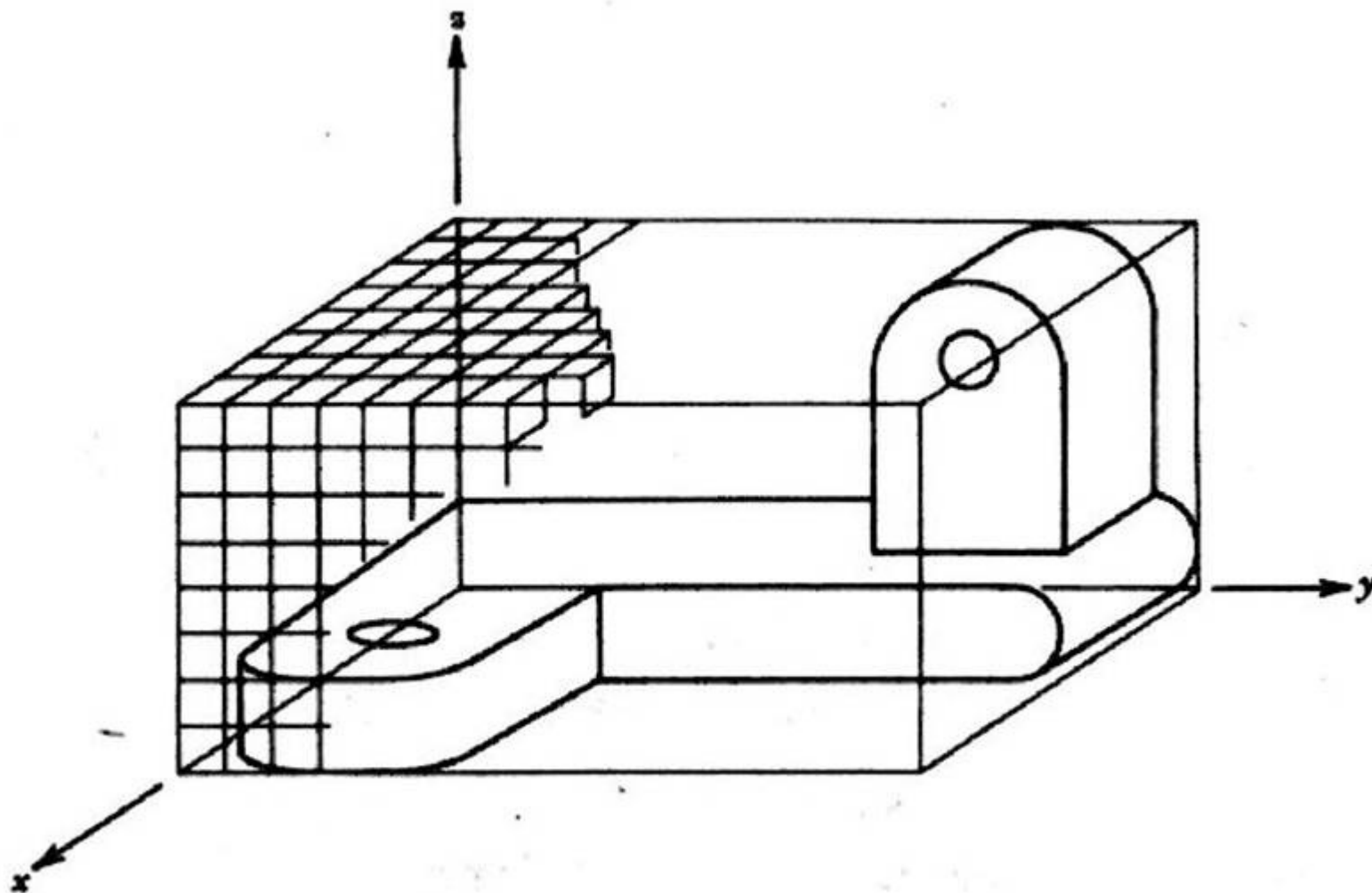


Parameterized primitive instantiation





Spatial occupancy enumeration





SOLID REPRESENTATION SCHEME

Spatial occupancy enumeration

This scheme is essentially a list of spatial cells occupied by the solid.

The cells, also called **voxels** are cubes of a fixed size and are arranged in a fixed spatial grid (other polyhedral arrangements are also possible but cubes are the simplest).

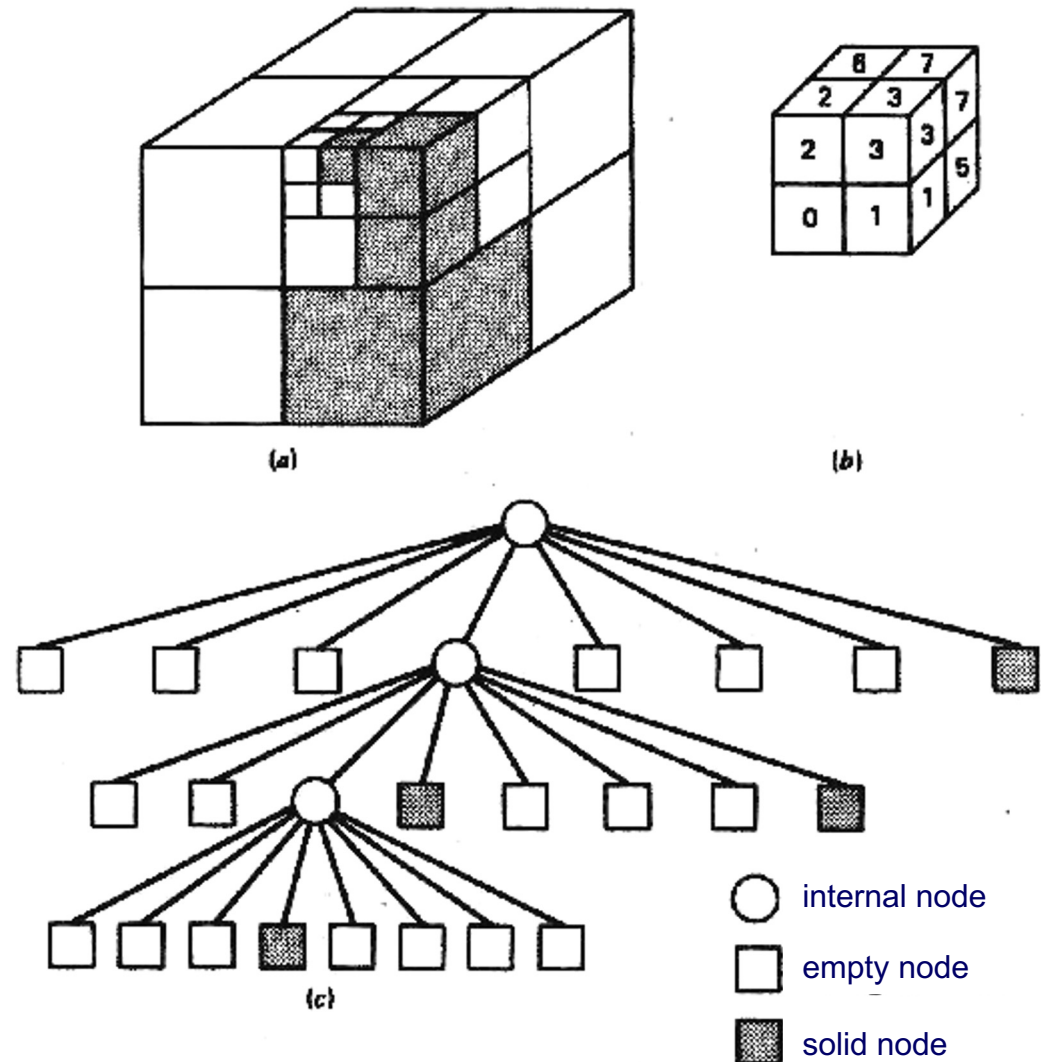
Each cell may be represented by the coordinates of a single point, such as the cell's centroid.

Spatial occupancy enumeration are unambiguous, but not unique solid representations and too verbose.



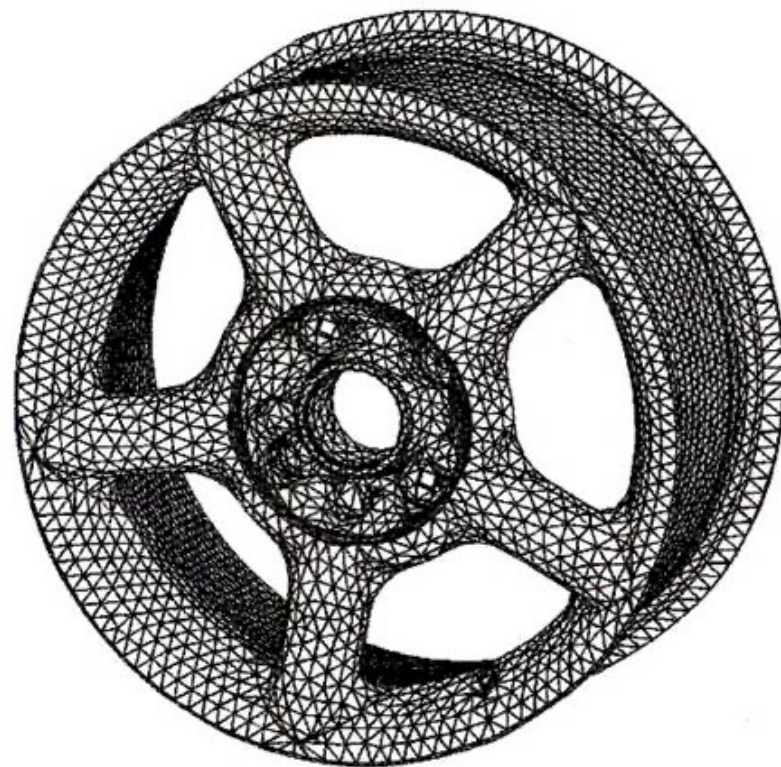
Octree

This spatial occupancy enumeration scheme is essentially a recursive subdivision of the 3D space in a tree data structure in which each internal node has exactly eight children (octants).





Cell Decomposition



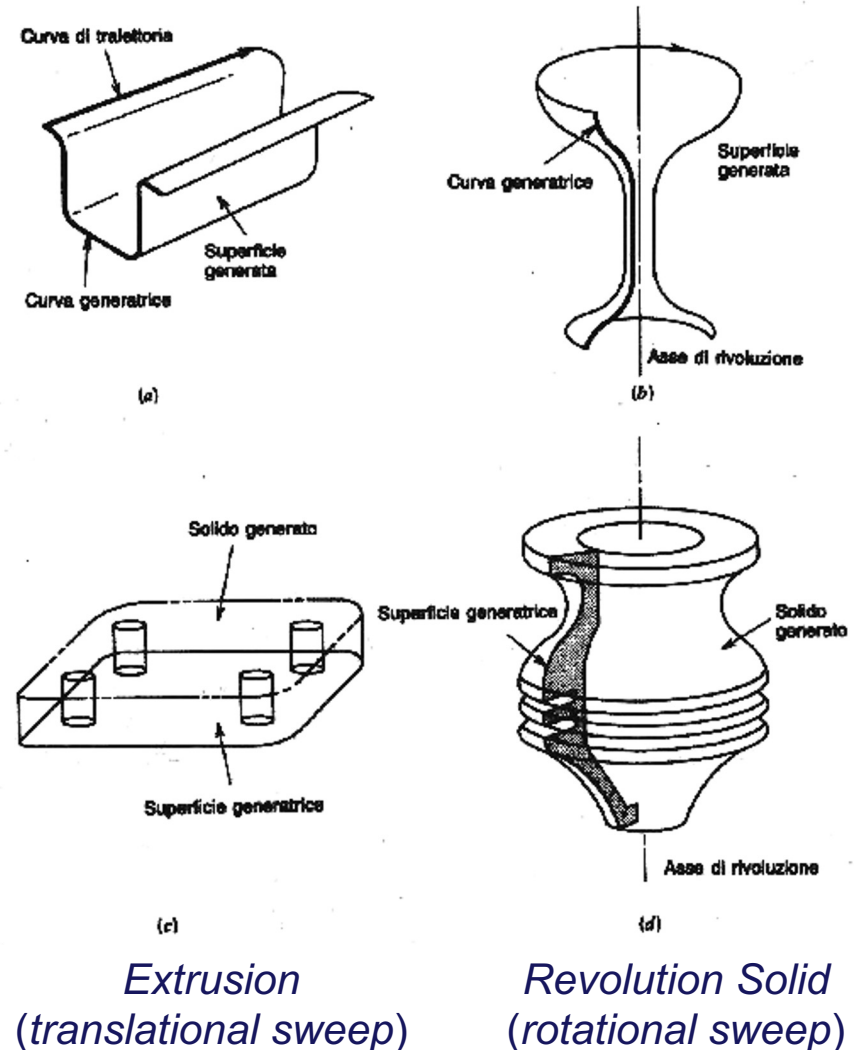


Sweeping

A point-set moving through space may trace or sweep out a line, a surface, or a volume (a solid) that may be represented by the moving point-set and its trajectory.

Required elements:

- *generatrix*: the moving point-set, it could be a point, a curve, a surface, or a solid;
- *directrix*: the trajectory.

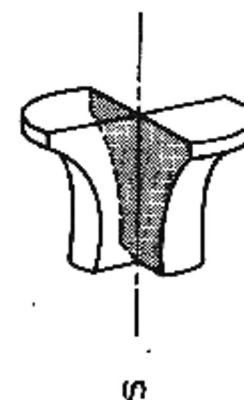
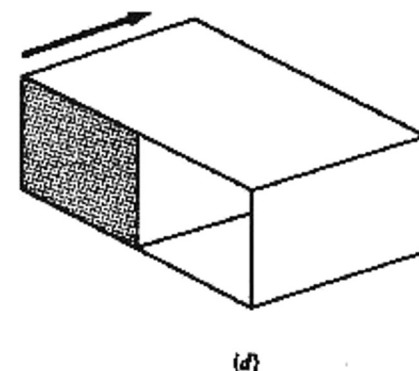
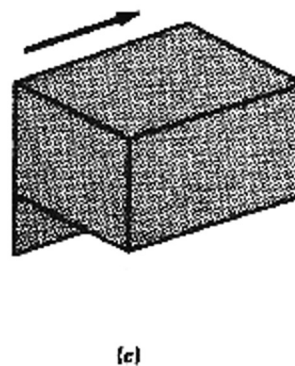
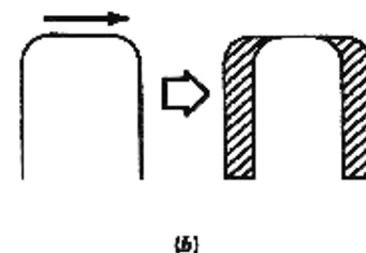
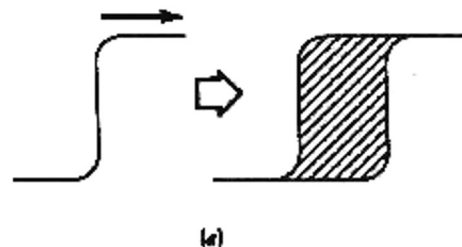




Errors in Sweeping

The wrong choice of generatrix and/or directrix may generate a non-manifold representation.

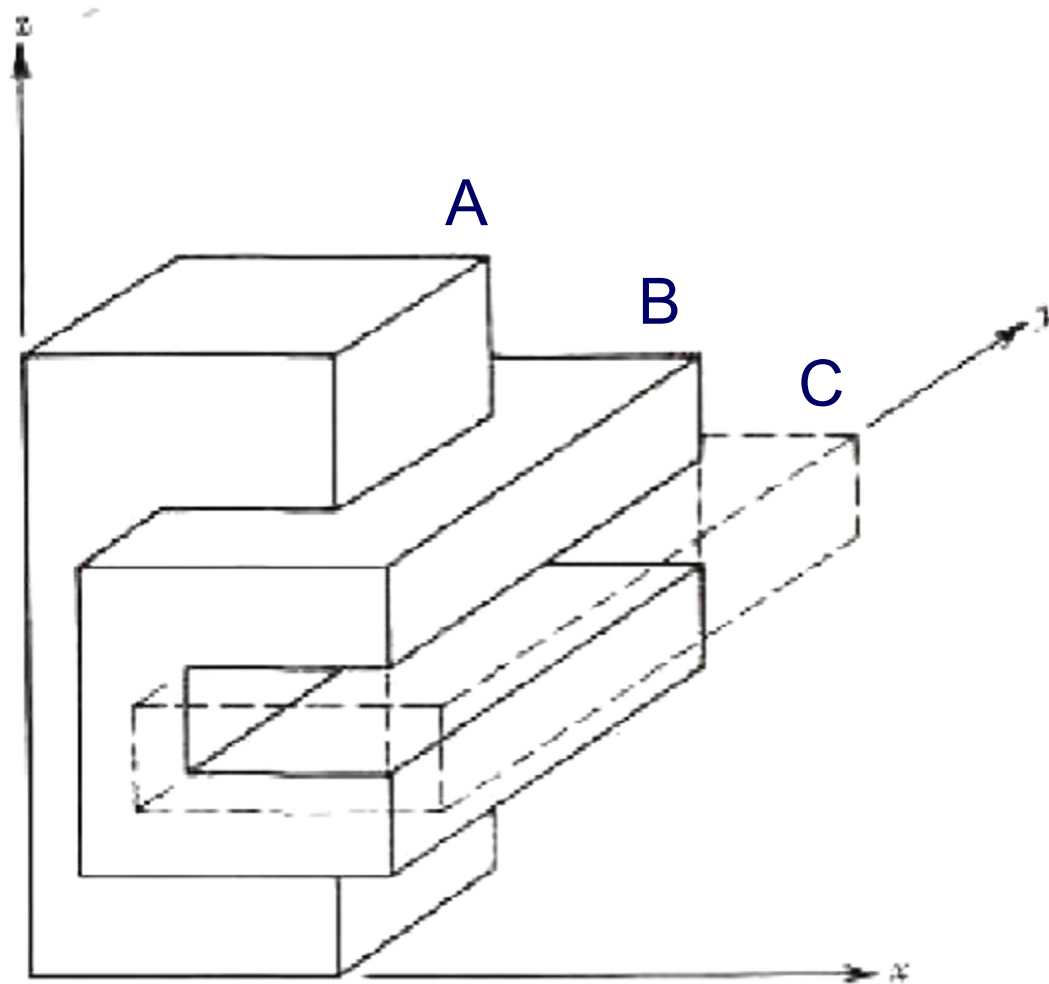
Homogeneous 3-dimensionality must be guaranteed, avoiding dangling edges or faces.





Constructive Solid Geometry (CSG)

It connotes a family of schemes for representing rigid solids as Boolean constructions or combinations of primitives via the regularized set operations.



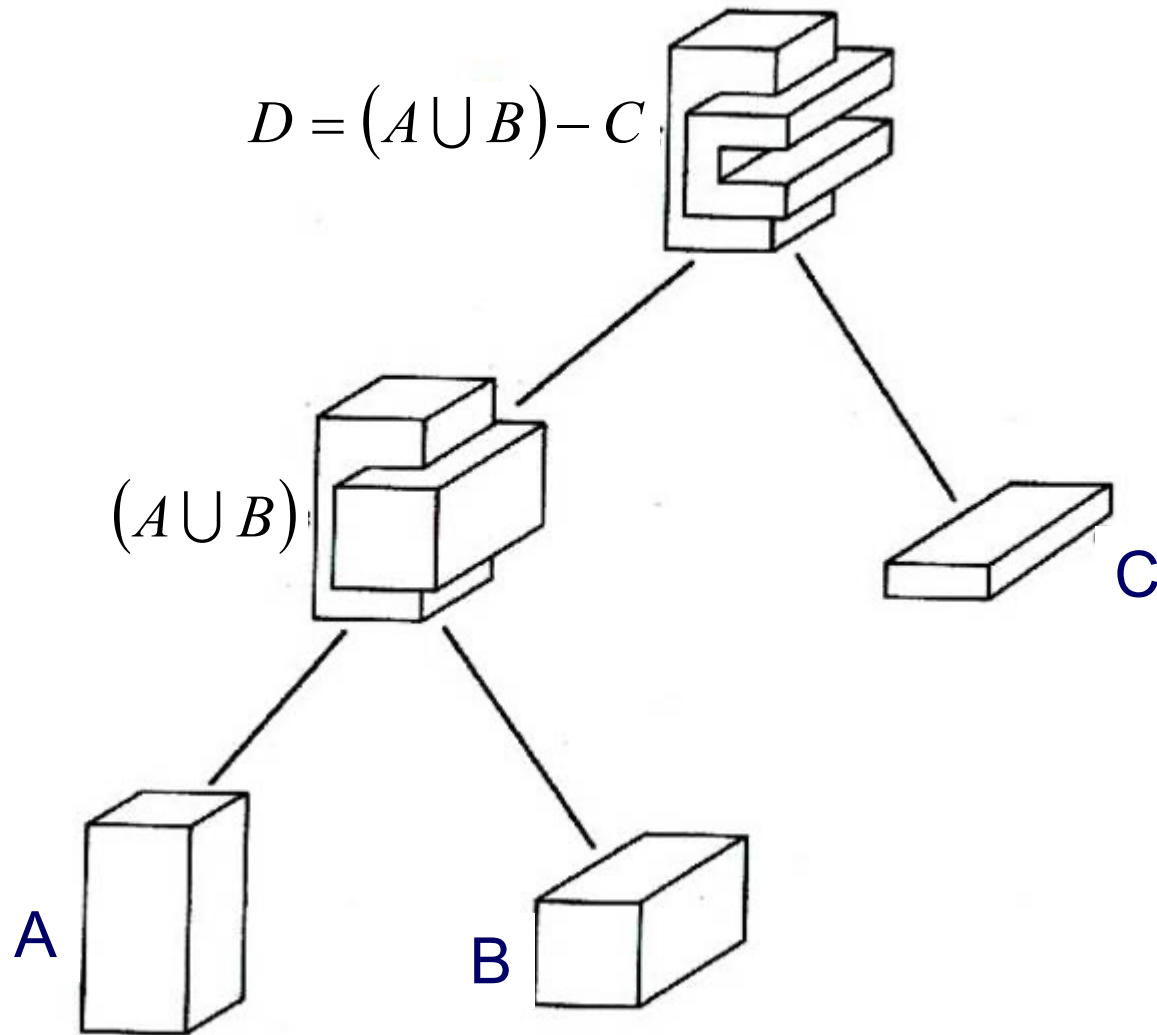
$$D = (A \cup B) - C$$

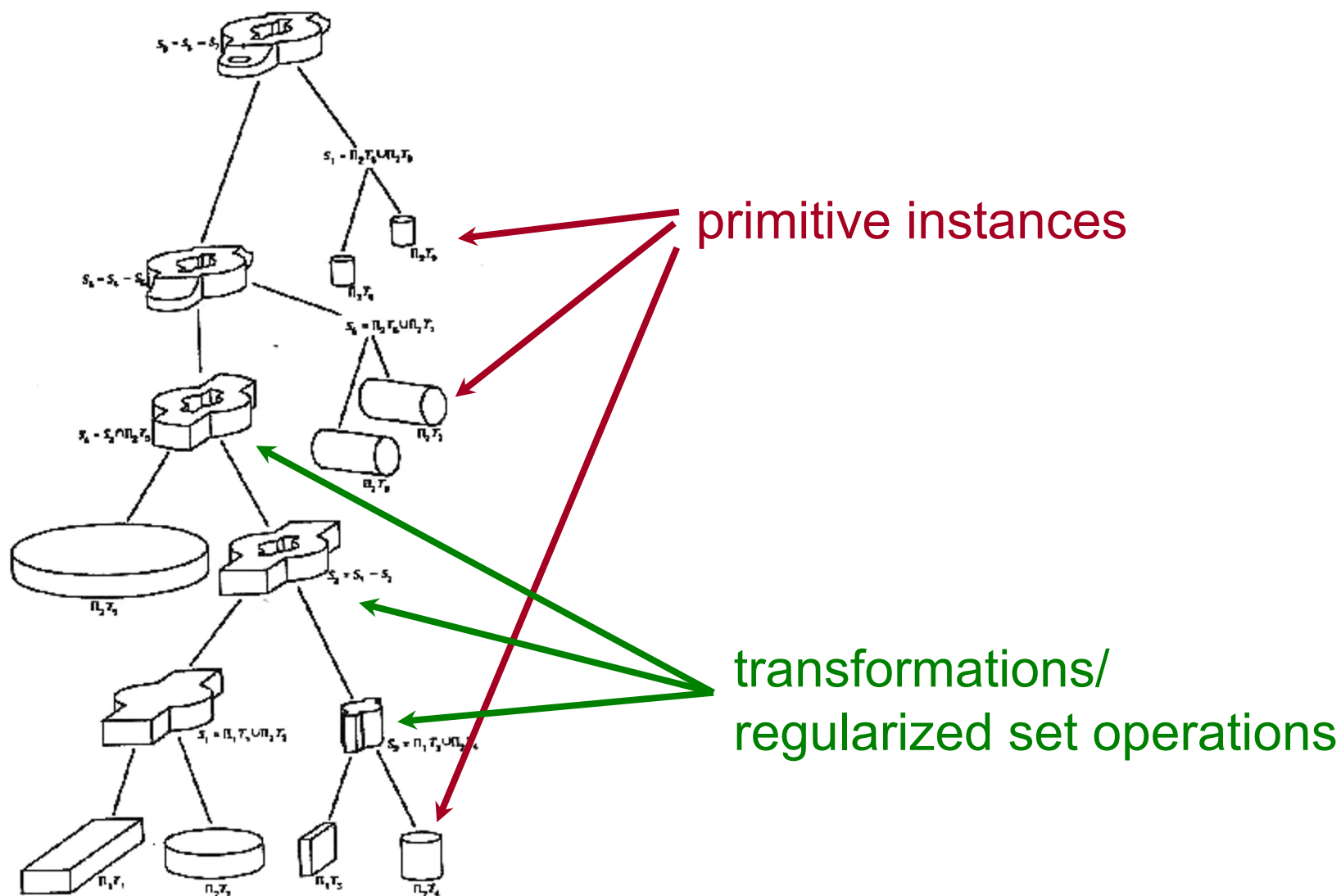


CSG representations take the form of ordered binary trees where non-terminal nodes represent either rigid transformations or regularized set operations.

Terminal nodes are primitive leaves that represent closed regular sets.

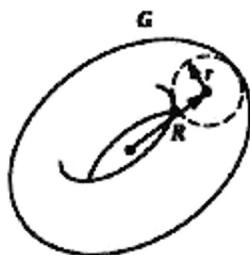
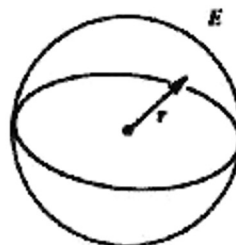
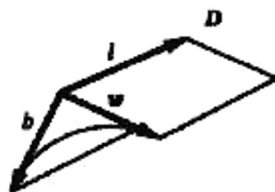
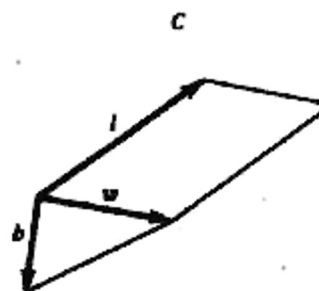
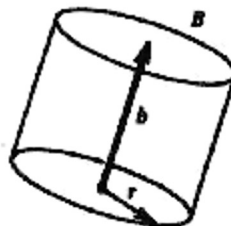
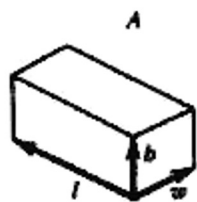
The semantics of CSG representations is clear. Each subtree represents a set resulting from applying the indicated transformations / regularized set operations on the set represented by the primitive leaves of the subtree.







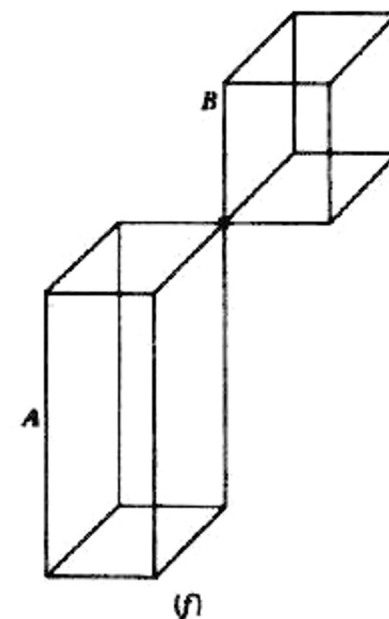
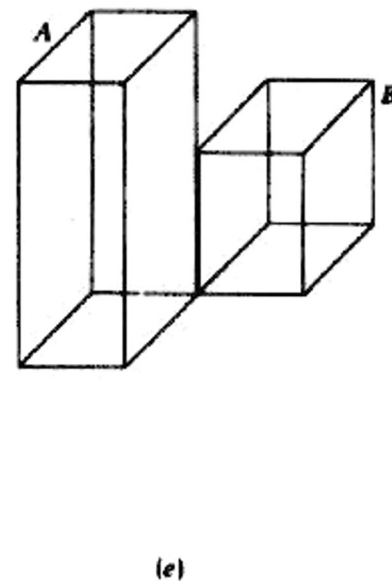
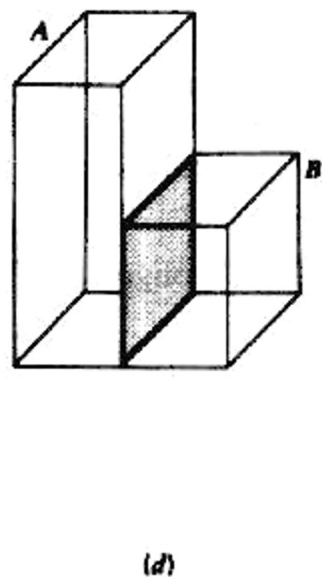
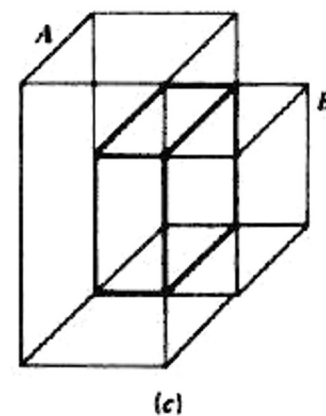
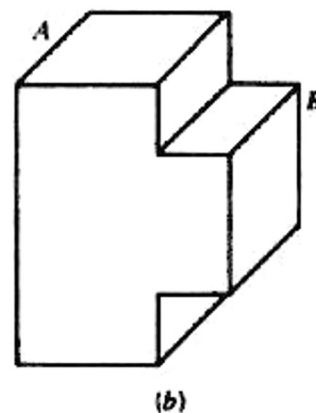
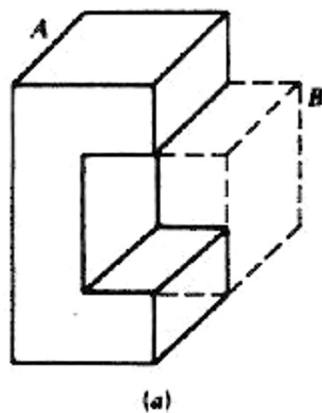
Primitive





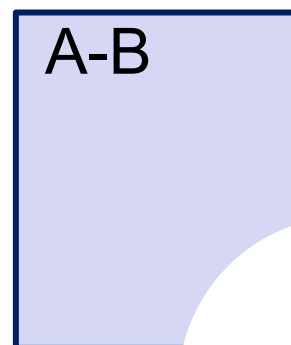
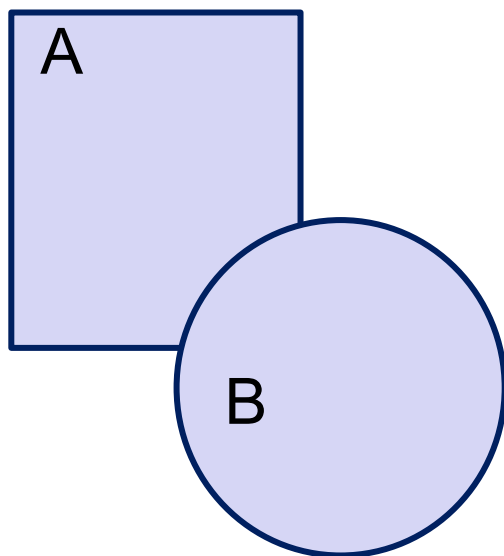
SOLID REPRESENTATION SCHEME

Boolean operations





Regularized Boolean operations



Boundary determinism

The boundary must contain the solid and hence determine the solid distinctly.



Regularized Boolean operations

$-^* \cup^* \cap^*$

The solid representation of an object is based on the idea that a physical object divides an 3-dimensional space R^3 into two regions: interior (i) and exterior separated by the object boundaries (b) (two-manifold objects).

Dimensional homogeneity is guaranteed for the class of closed regular sets, defined as sets equal to the closure (k) of their interior: $S = kiS = iS \cup bS$.

$$A -^* B = ki(A - B)$$

$$A \cup^* B = ki(A \cup B)$$

$$A \cap^* B = ki(A \cap B)$$

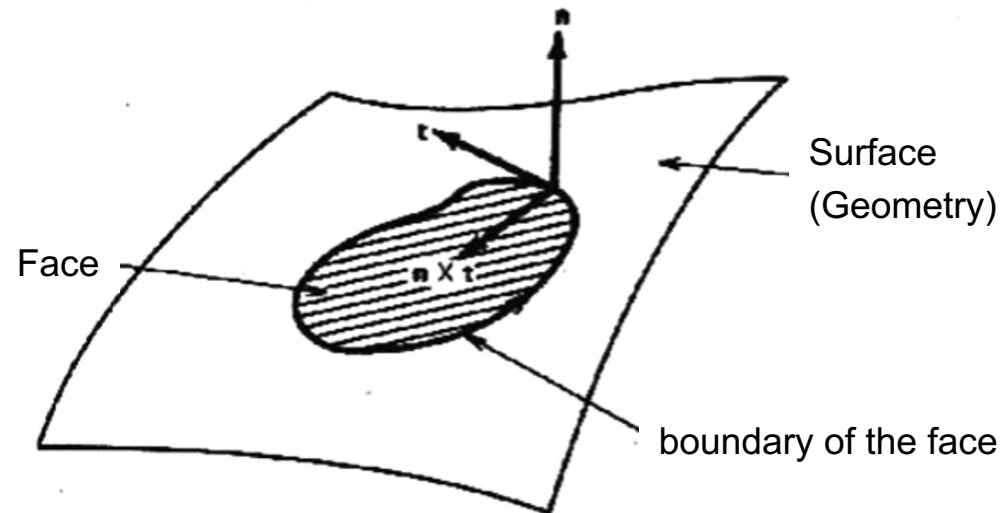
Boundary Representation - BRep

A solid is represented as a collection of connected surface elements, the boundary between solid and non-solid.

Boundary representation of models are composed of two parts: topology and geometry (surfaces, curves, and points).

The main topological items are: faces, edges, and vertices.

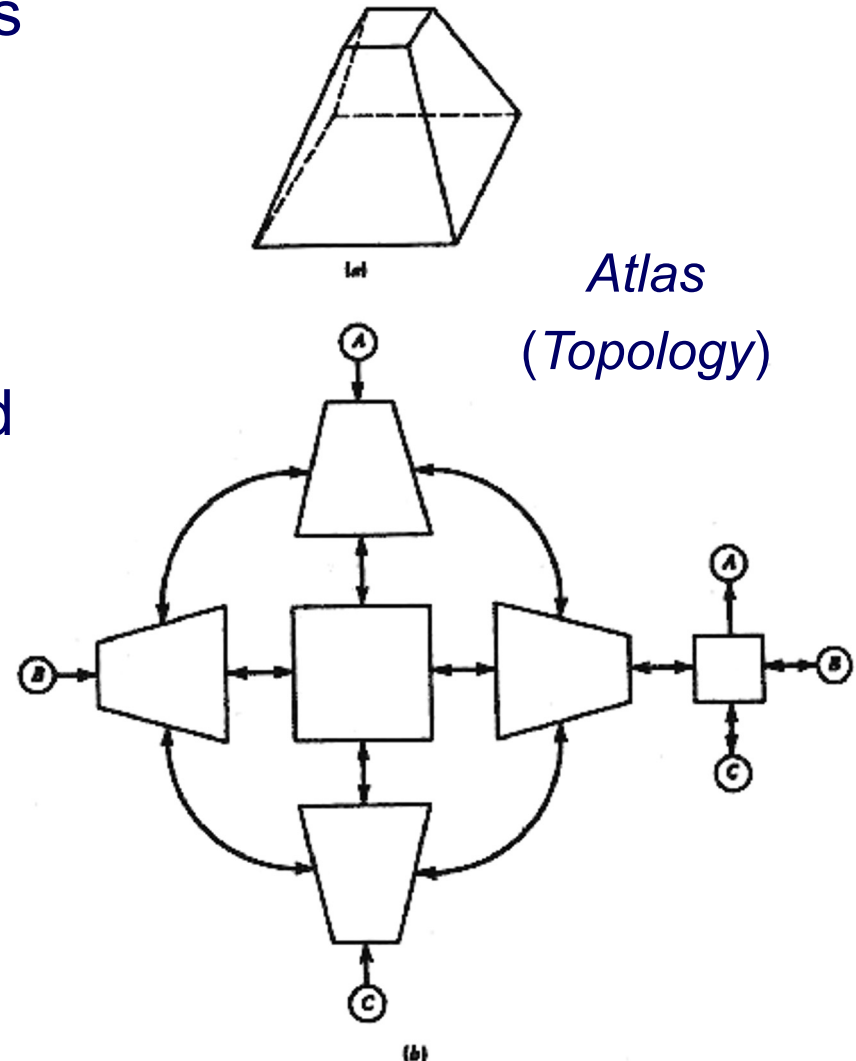
A face is a bounded portion of a surface, an edge is a bounded piece of a curve, and a vertex lies at a point.





The topological relationships are a specific subset of the large variety of spatial relationships.

They are characterized by the property to be preserved under topological transformations, such as translation, rotation, and scaling.

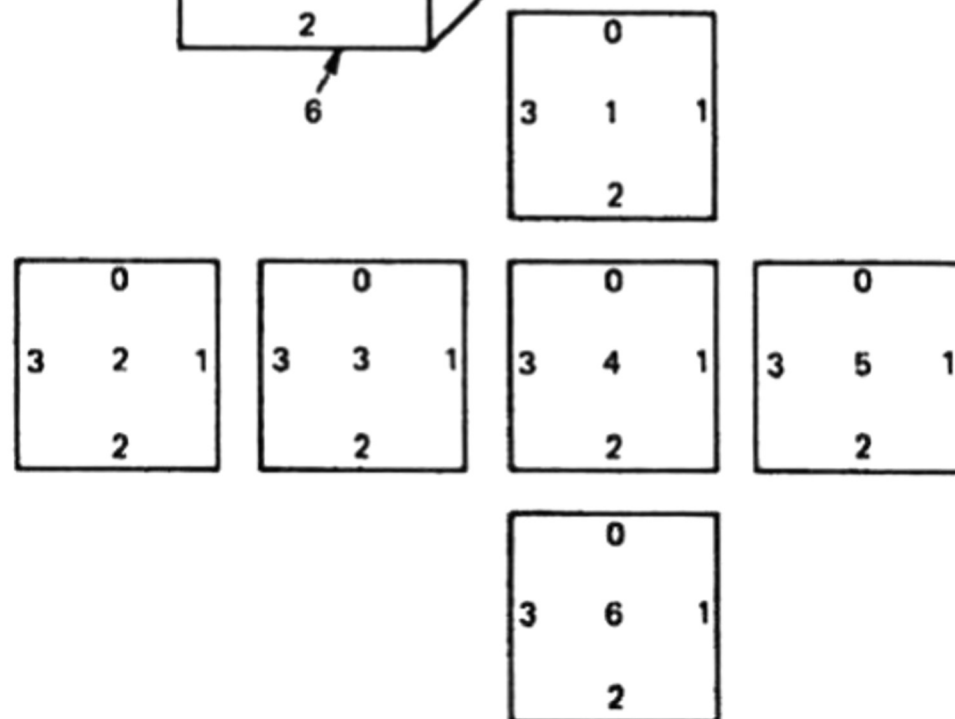
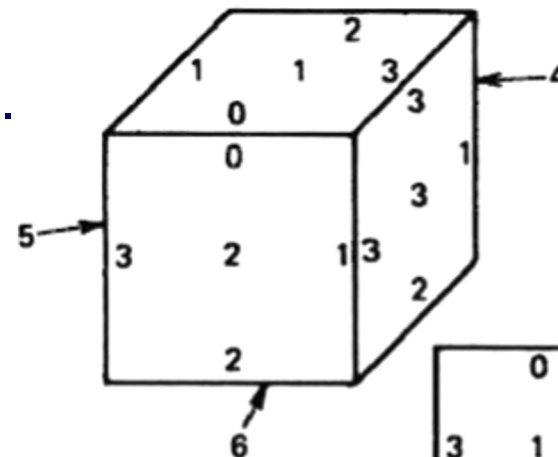




An example of topological relationship is the adjacency matrix.

E.g. if we consider the 6 faces of the cube, the following matrix is valid:

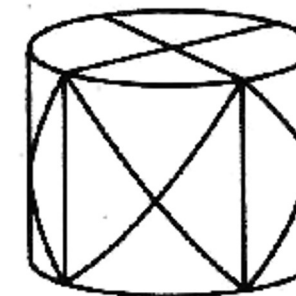
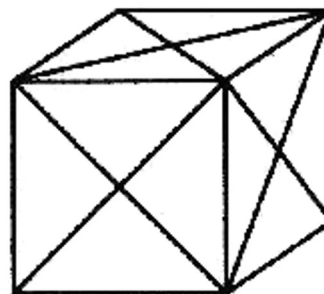
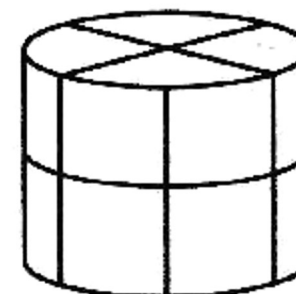
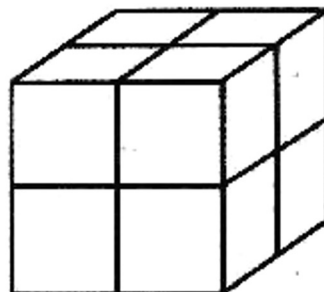
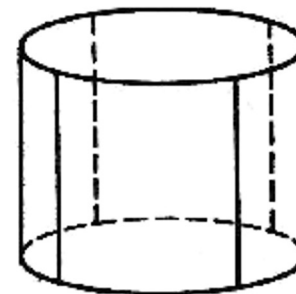
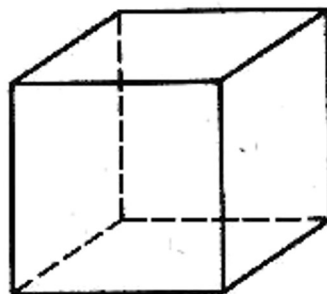
	1	2	3	4	5	6
1	0	1	1	1	1	0
2	1	0	1	0	1	1
3	1	1	0	1	0	1
4	1	0	1	0	1	1
5	1	1	0	1	0	1
6	0	1	1	1	1	0





Boundary Representation:

- valid;
- unambiguous;
- not unique.

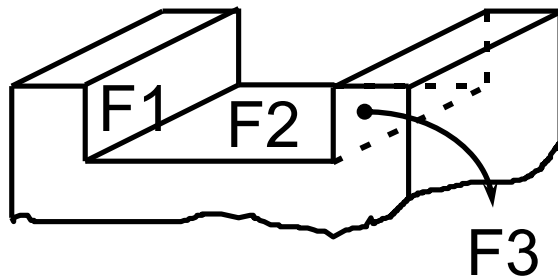




COMPUTER-COMPATIBLE PART DESCRIPTION

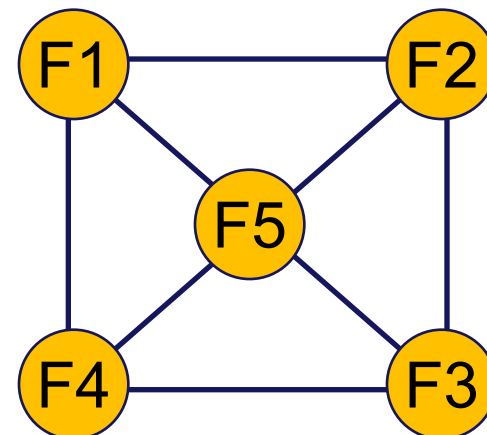
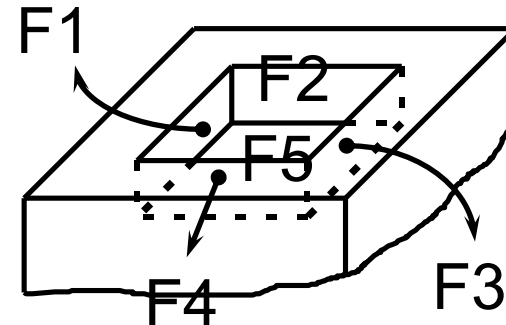
- Form Feature and Form Feature Recognition

Slot



— Concave Edge

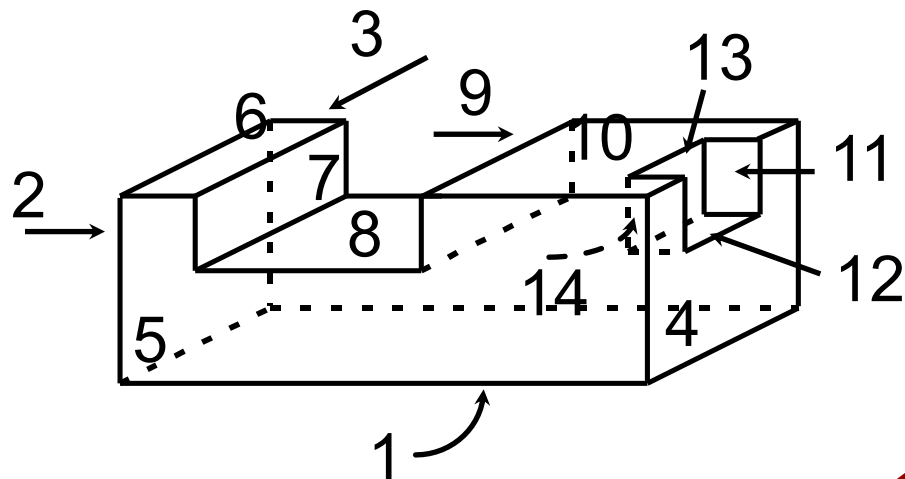
Pocket





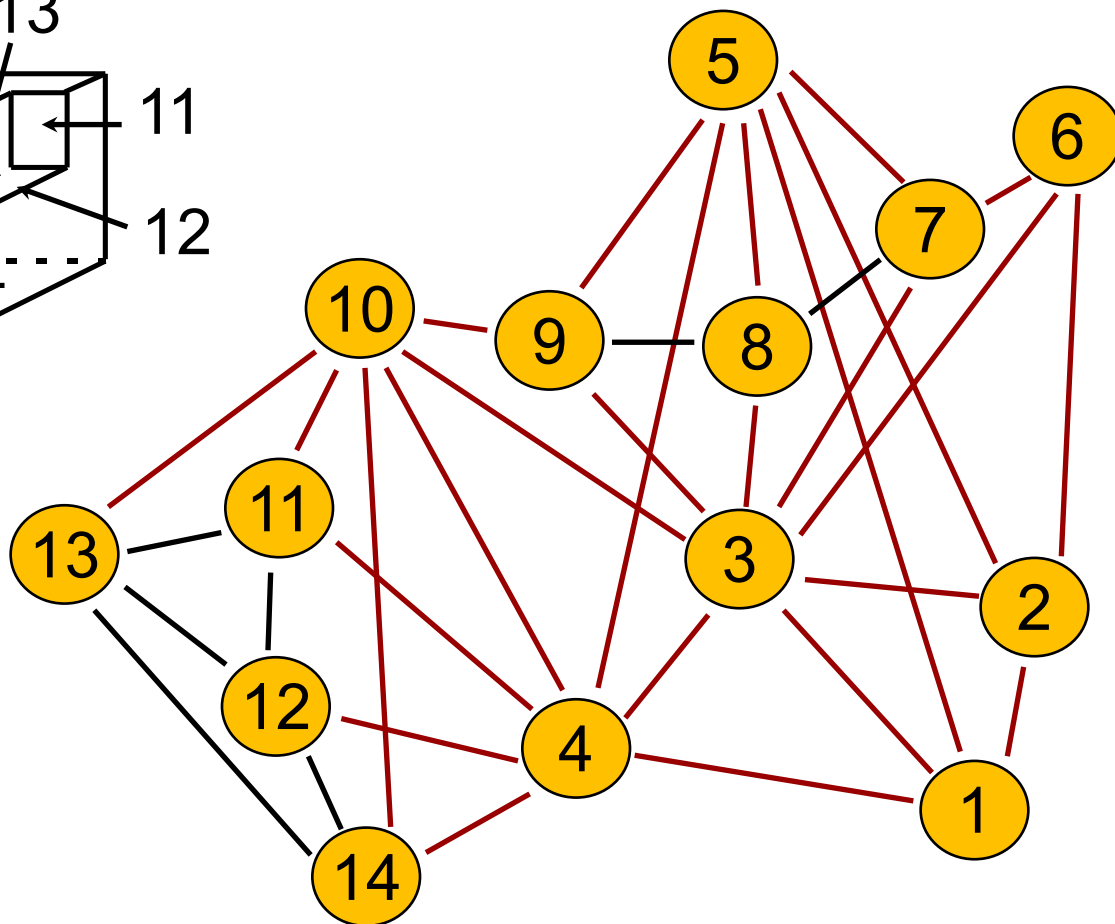
COMPUTER-COMPATIBLE PART DESCRIPTION

Form Feature and Form Feature Recognition



— Concave Edge
— Convex Edge

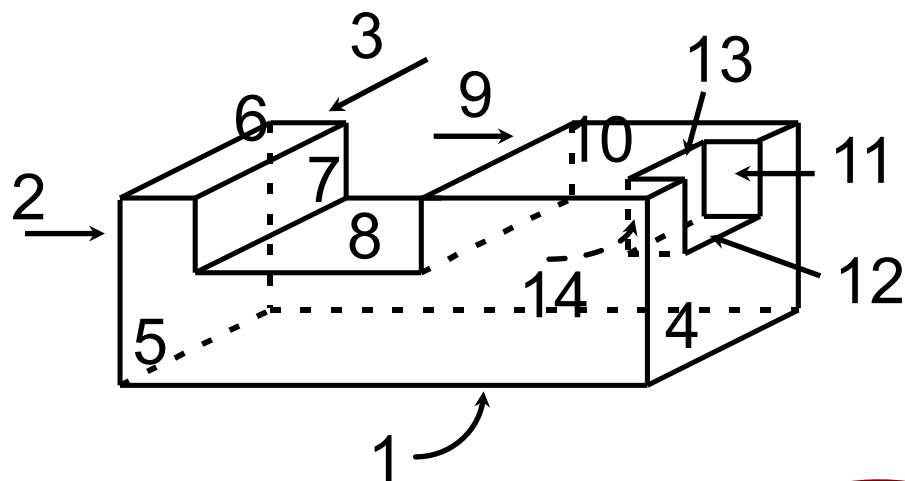
Boundary Representation Graph





COMPUTER-COMPATIBLE PART DESCRIPTION

Form Feature and Form Feature Recognition



Boundary Representation Graph

— Concave Edge
— Convex Edge

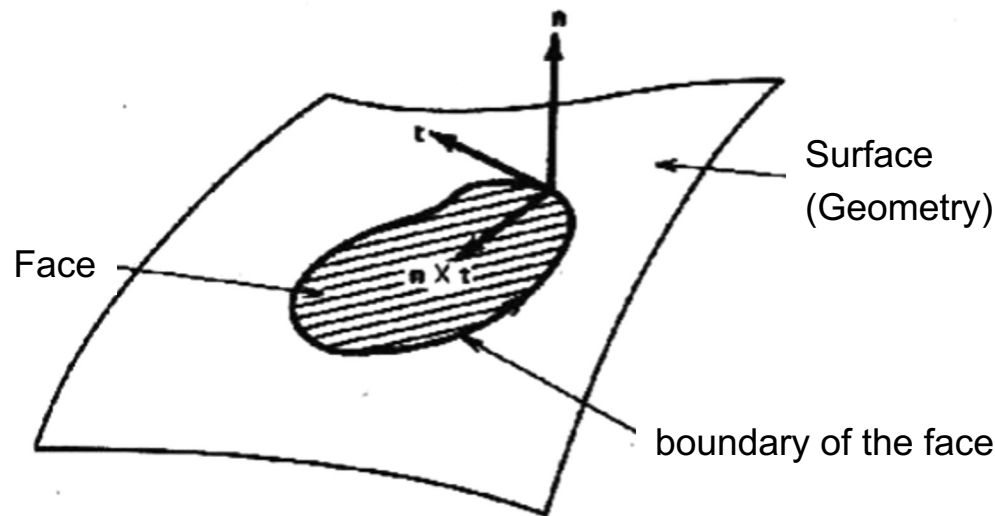
Open pocket

Slot

GEOMETRIC MODELING

Boundary representation of models are composed of two parts: **topology and geometry** (surfaces, curves, and points).

A face is a bounded portion of a surface, an edge is a bounded piece of a curve, and a vertex lies at a point.





A polynomial curve has the following parametric equation:

$$P(t) = t^n a_n + t^{n-1} a_{n-1} + t^{n-2} a_{n-2} + \cdots + a_0$$

n is the degree of the curve

The order of the curve is equal to the number of coefficients.
The following relationship is always valid: order = degree + 1

E.g.:

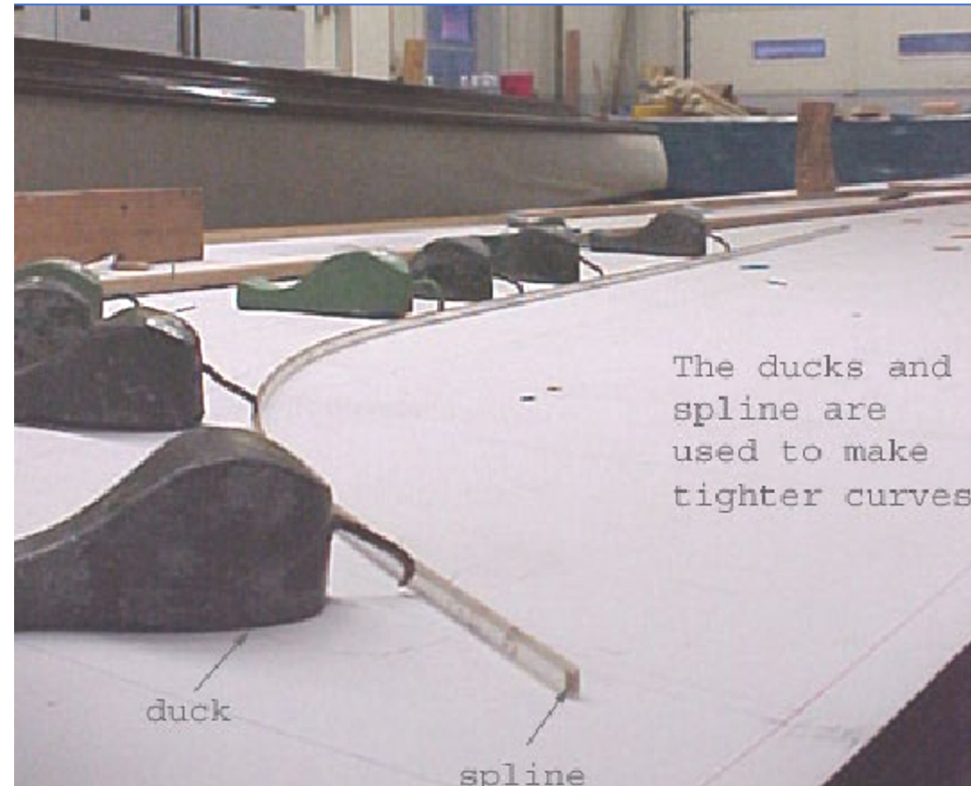
$$P(t) = t a_1 + a_0$$

$$P(t) = t^2 a_2 + t a_1 + a_0$$

$$P(t) = t^3 a_3 + t^2 a_2 + t a_1 + a_0$$



They derive their name from an instrument used for the technical drawing called "spline" consisting of a wooden or plastic rod, which is flexed so to pass exactly through a series of points, with a gradual change of curvature.





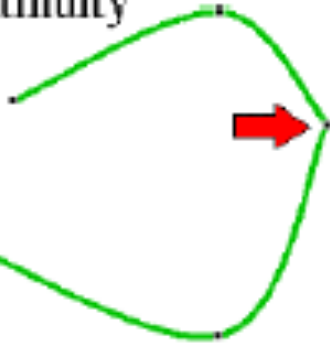
Splines are **interpolating functions** which, starting from a certain number of points (called **knots**), connect each pair of successive knots with a polynomial function. In order for the entire curve to be **smooth**, the functions must connect in an proper way.

Definition: A spline of degree k and order $k + 1$ is a function $S: [a, b] \rightarrow R$ for which there exist real numbers $x_0, \dots, x_n$, with $a = x_0 < x_1 < \dots < x_n = b$ so that:

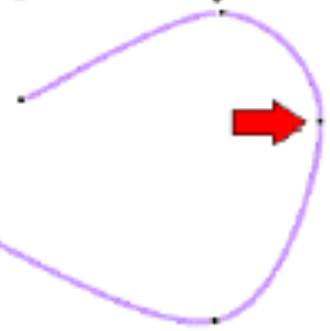
- (i) $s_{[x_j, x_{j+1}]}^k \in P^k, \quad j = 0, 1, \dots, n-1$
- (ii) $s^k \in C^{k-1} \quad [a, b]$



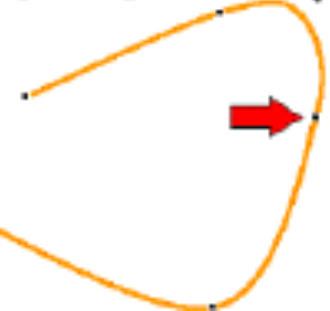
C_0 continuity



C_0 & C_1 continuity



C_0 & C_1 & C_2 continuity



Continuity

C^0 : all continuous functions.

C^1 : all differentiable functions whose derivative is continuous, such functions are called continuously differentiable.

C^2 : all differentiable functions whose second derivative is continuous.

C^k continuity or “*k-smoothness*” for any positive integer k to be the set of all differentiable functions whose derivative is in C^{k-1}



Definition: A spline $S: [a, b] \rightarrow R$ for which there exist real numbers $x_0, \dots, x_n$, with $a = x_0 < x_1 < \dots < x_n = b$ is a cubic spline if $k = 3$.

$$(i) \quad s_{[x_j, x_{j+1}]}^3 \in P^3, \quad j = 0, 1, \dots, n-1$$

$$(ii) \quad s^3 \in C^2 \quad [a, b]$$

$$s_j(x) = a_{3,j} \cdot x^3 + a_{2,j} \cdot x^2 + a_{1,j} \cdot x + a_{0,j}$$



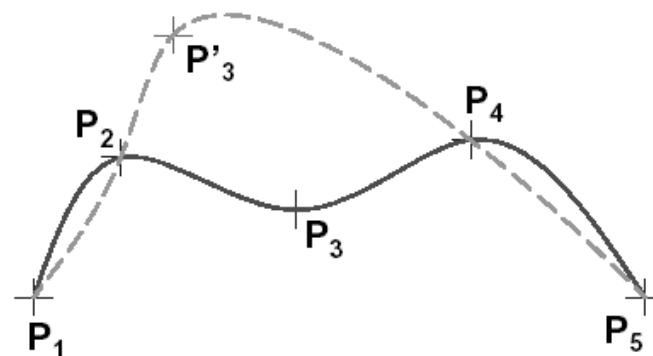
In the case of $n + 1$ knots points, there are n intervals, so in the case of a cubic spline we must calculate $4 n$ unknowns.

The conditions to be imposed are:

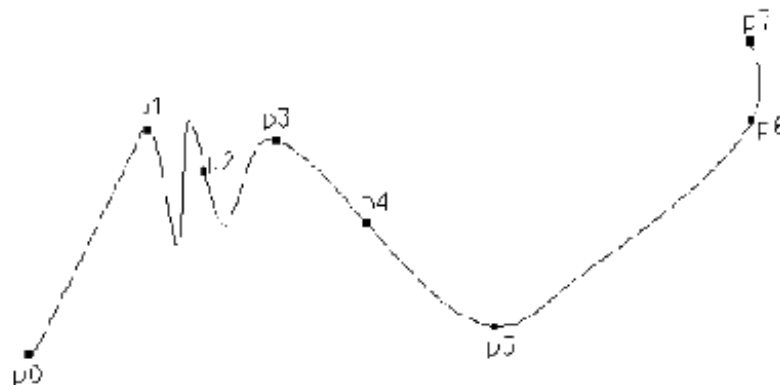
- passage through the knots,
- continuity of the first derivative,
- continuity of the second derivative
- second derivatives equal to zero at the external nodes (natural splines).



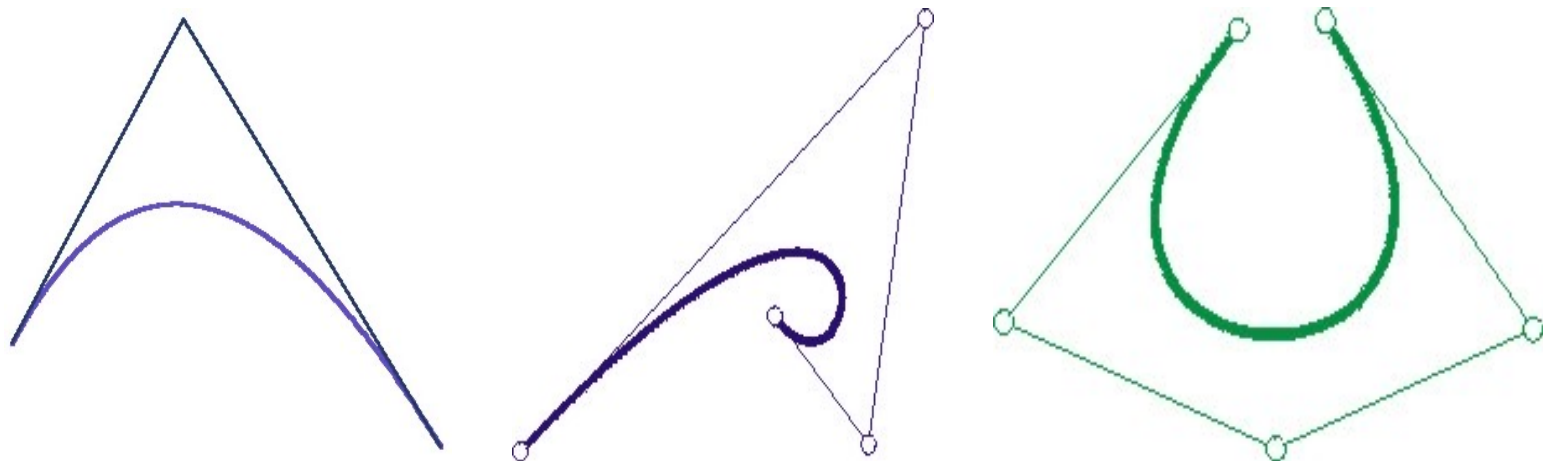
Moving a point to interpolate changes the entire curve.



Interpolating non-equidistant points creates unwanted fluctuations.



- Bézier curves have been developed from the need for freeform curves in CAD/CAM/CAE technology
- P. de Casteljau (1959) at Citroën, P. Bézier (1962) at Renault
- Bézier curves are contained in almost all CAD systems
- The shape of a Bézier curve is manipulated by a control polygon, whose vertices are called control points





BÉZIER CURVE: DE CASTELJAU ALGORITHM

Insertion of points according to a fixed ratio yields a sequence of polygons with decreasing number of vertices, until a point of the desired curve remains.

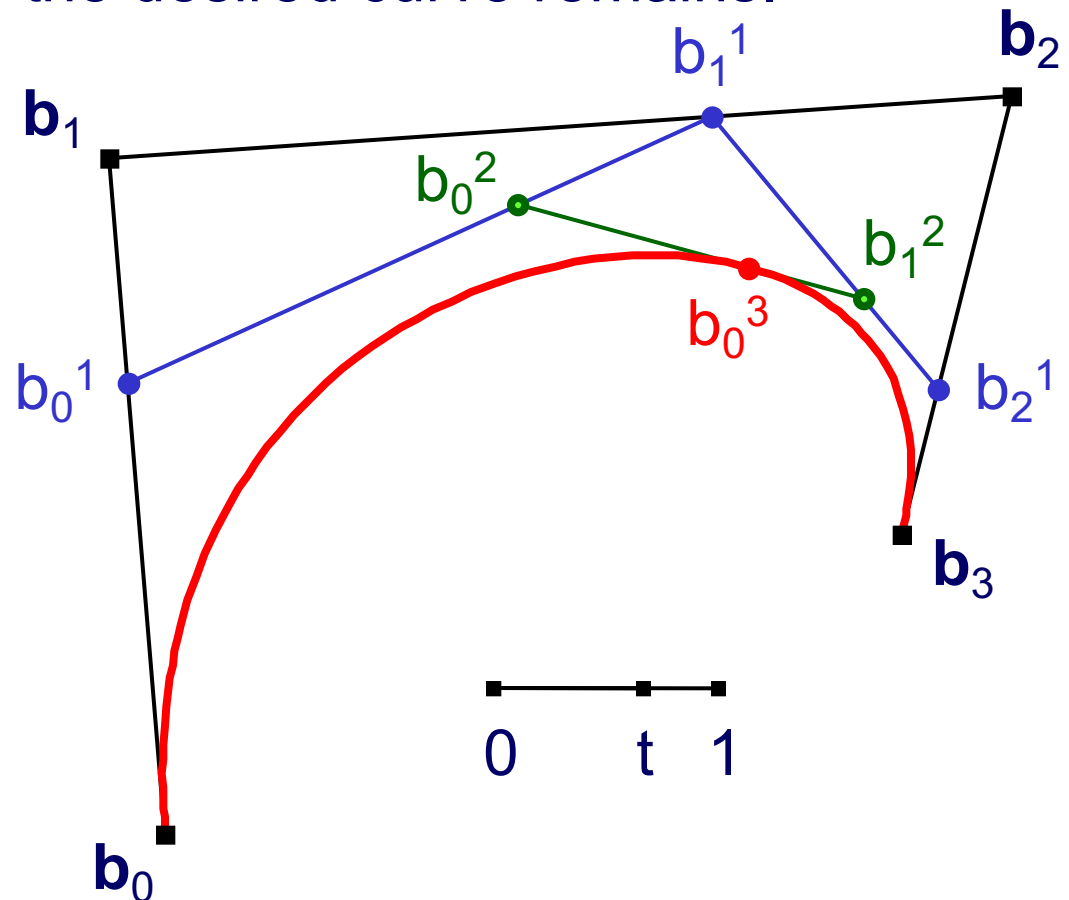
Given: control points
 $b_0, \dots, b_n$

Find: Points on the Bezier curve of degree n

Each point has a parameter t from the interval $[0,1]$:

$t = 0$ corresponds to b_0

$t = 1$ corresponds to b_n





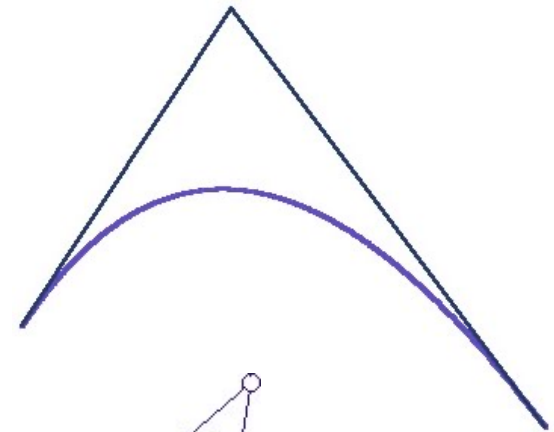
Degree 1

- **linear** Bézier curve



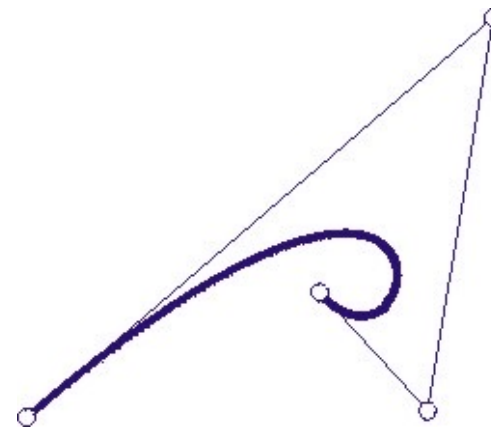
Degree 2

- **quadratic** Bézier curve



Degree 3

- **cubic** Bézier curve





If we consider two control points (P_0, P_1) , the Bézier curve is:

$$P_0 \cdot (1-t) + P_1 \cdot t$$

If we consider three control points (P_0, P_1, P_2) , the Bézier curve is:

$$P_0 \cdot (1-t)^2 + P_1 \cdot 2(1-t)t + P_2 \cdot t^2$$

If we consider four control points (P_0, P_1, P_2, P_3) , the Bézier curve is:

$$P_0 \cdot (1-t)^3 + P_1 \cdot 3(1-t)^2t + P_2 \cdot 3(1-t)t^2 + P_3 \cdot t^3$$



BERNSTEIN POLYNOMIAL APPROXIMATION

If we consider four control points (P_0, P_1, P_2, P_3) , the Bézier curve is:

$$P_0 \cdot (1-t)^3 + P_1 \cdot 3(1-t)^2t + P_2 \cdot 3(1-t)t^2 + P_3 \cdot t^3$$

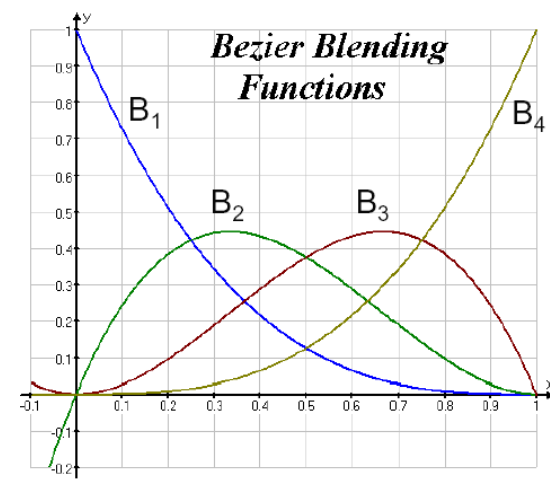
It's a function with four coefficients called Bernstein polynomials:

- $B_{30}(t) = (1-t)^3$
- $B_{31}(t) = 3(1-t)^2t$
- $B_{32}(t) = 3(1-t)t^2$
- $B_{33}(t) = t^3$

$$B_{n,i}(t) = \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i}$$



$$P(t) = \sum_{i=0}^3 B_{3,i}(t) P_i$$





If we consider n control points $(P_0, P_1, \dots, P_n)$,
the Bézier curve is:

$$P(t) = \sum_{i=0}^n B_{n,i}(t) P_i \quad 0 \leq t \leq 1$$

P_i : *control point*

$\{P_i\}$: *control polygon*

n : *degree of the curve*

$n+1$: *order of the curve*

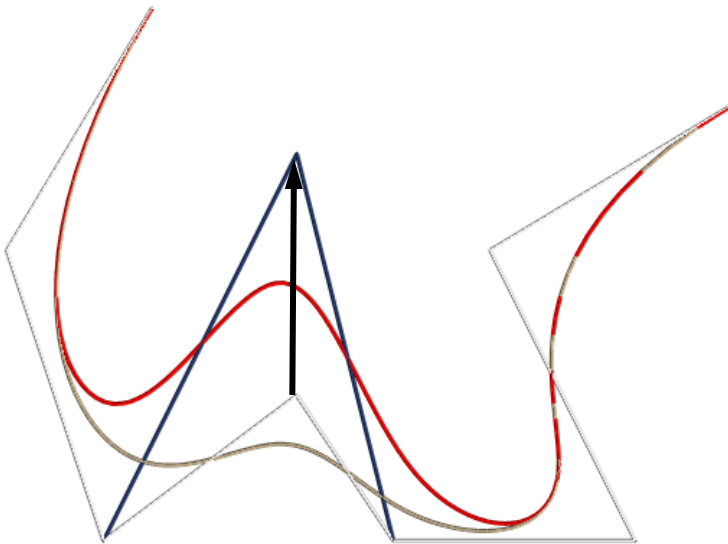
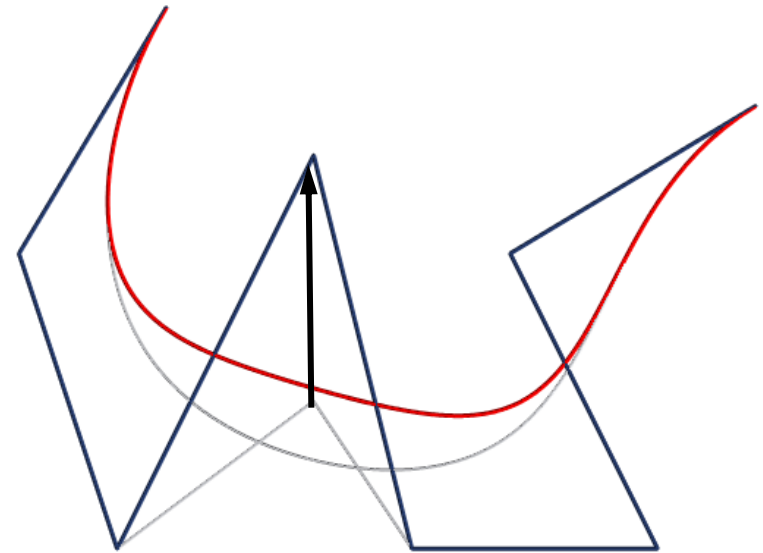
$$B_{n,i}(t) = \frac{n!}{i!(n-i)!} t^i (1-t)^{n-i}$$

Bernstein polynomials are the
Bézier blending functions.



Bézier curves are completely determined by their control polygon. Changing a single control point results in a change of the entire curve (global).

⇒ Not well-suited for design purposes



A possible solution: Curves of low degree are stitched together to a more complicated curve
⇒ *spline curve*, locally controllable, at segment joints one needs appropriate transition conditions (e.g. common tangent)



$$\mathbf{P}(t) = \mathbf{P}_0 \cdot f_0(t) + \mathbf{P}_1 \cdot f_1(t) + \dots + \mathbf{P}_n \cdot f_n(t)$$

If the blending function is zero for a particular value of t , then the corresponding control points do not affect the curve at that particular value of t .

Therefore, to solve the problem of local control, we consider blending functions that are always null except for certain values of t .



Given:

$n + 1$ control points $P_0, P_1, P_2, \dots, P_n$

$m + 1$ knots $t_0, t_1, t_2, \dots, t_m$

a B-spline curve is a polynomial curve defined as:

$$P(t) = \sum_{i=0}^n N_{i,k}(t) P_i$$

- $P_i : n+1$ control points
- $\{P_i : i = 0, 1, \dots, n\}$: control polygon
- k : degree of the curve
- $k+1$: order of the curve



B-SPLINE CURVE: BLENDING FUNCTIONS

Given $n + 1$ control points $P_0, P_1, P_2, \dots, P_n$

$m + 1$ knots $t_0, t_1, t_2, \dots, t_m$

a B-spline blending functions are defined as follows:

If $k = 0$

$$N_{i,0}(t) = \begin{cases} 1 & \text{if } t \in [t_i, t_{i+1}] \\ 0 & \text{if } t \notin [t_i, t_{i+1}] \end{cases}$$

If $k > 0$

$$N_{i,k}(t) = \frac{t - t_i}{t_{i+k} - t_i} N_{i,k-1}(t) + \frac{t_{i+k+1} - t}{t_{i+k+1} - t_{i+1}} N_{i+1,k-1}(t)$$

convention: $0/0 = 1$

It can be proof that:

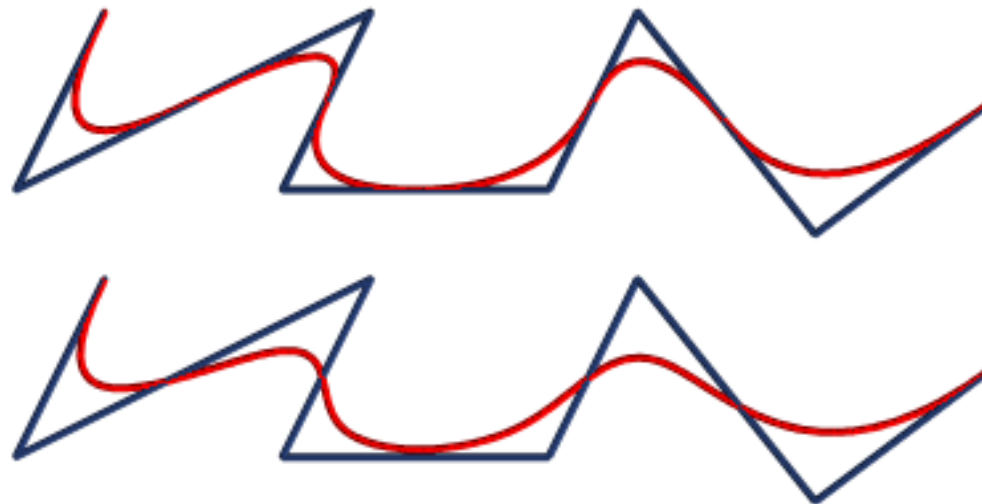
$$\sum_{i=0}^n N_{i,k}(t) = 1$$



A B-spline curve of degree d consists of Bezier curves of degree d , which are put together with optimal smoothness:

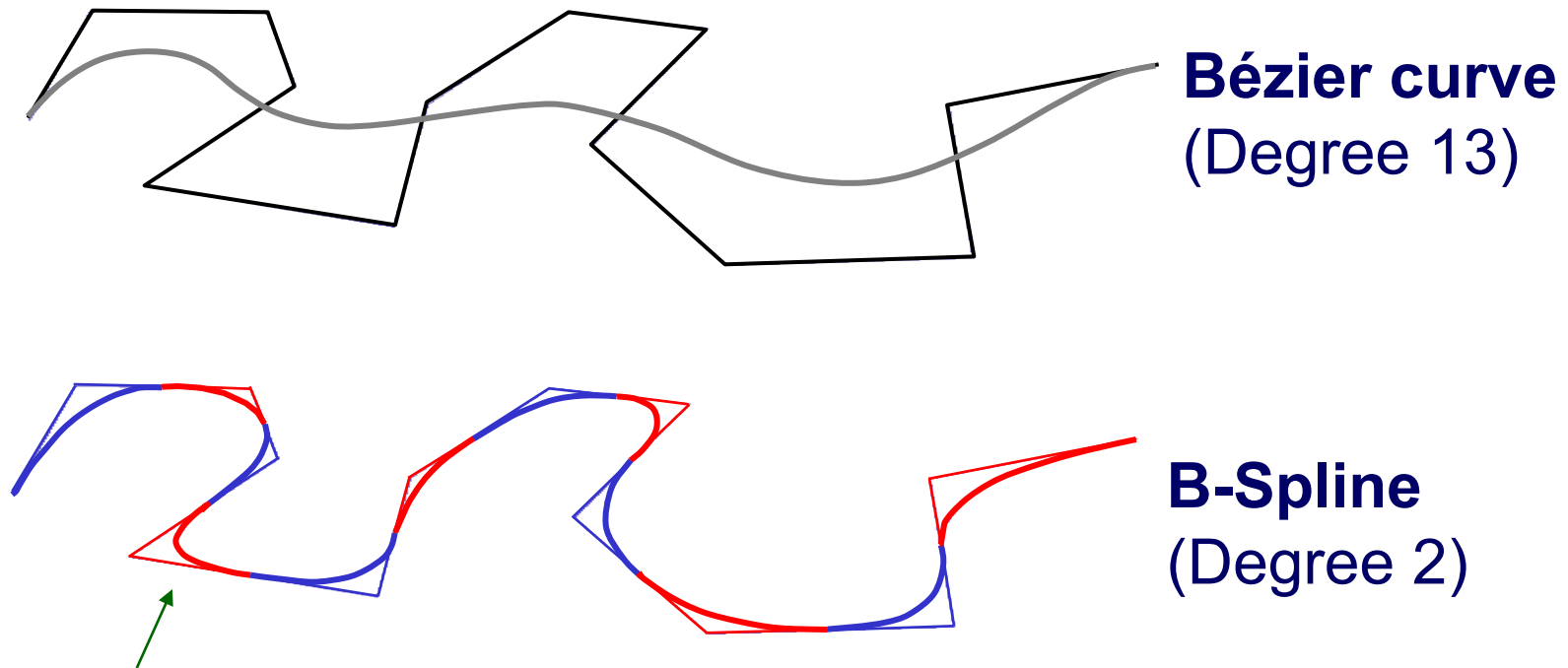
- degree 2: continuous tangent
- degree 3: continuous curvature

Input: control polygon, degree, knots.





Example: two curves with the same control polygon

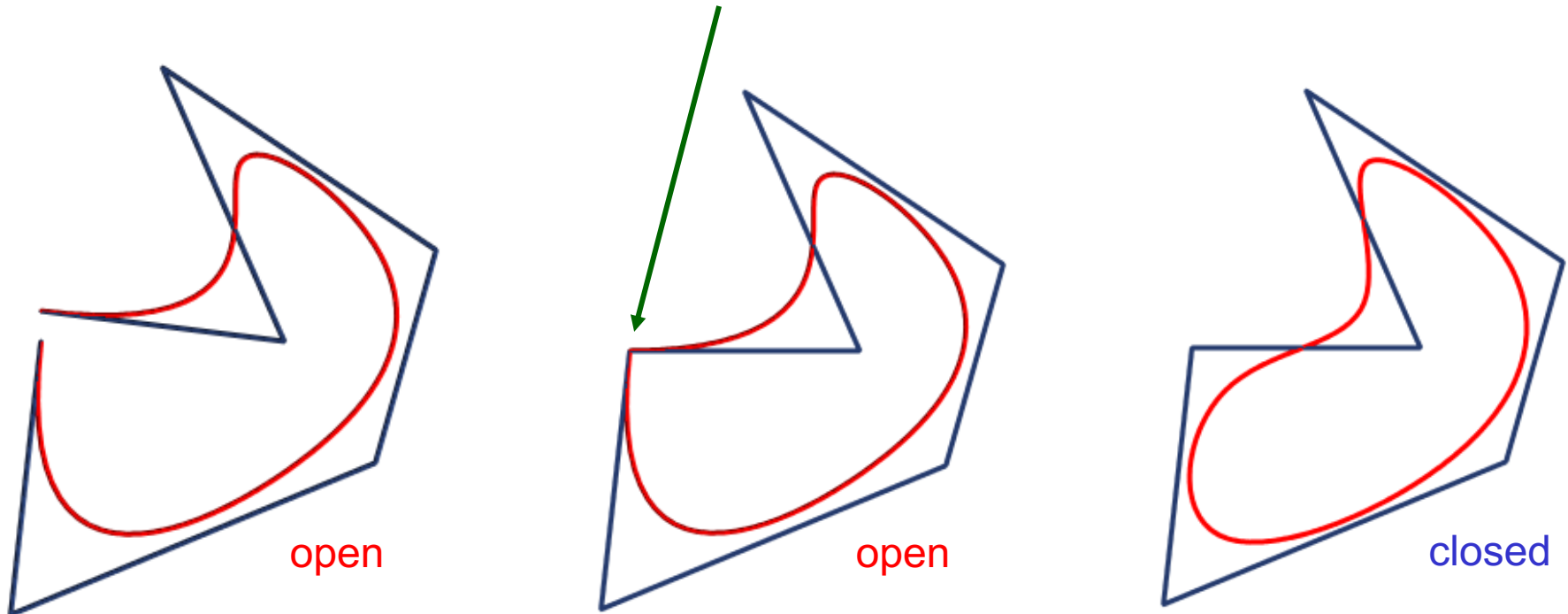


Parabola segments joined with tangent continuity



B-spline curves can be *open* or *closed*:

- For a closed B-spline curve a closed control polygon is entirely smoothed
- In open mode, even a closed polygon has an initial (and identical) end point; there we have no smoothing effect.

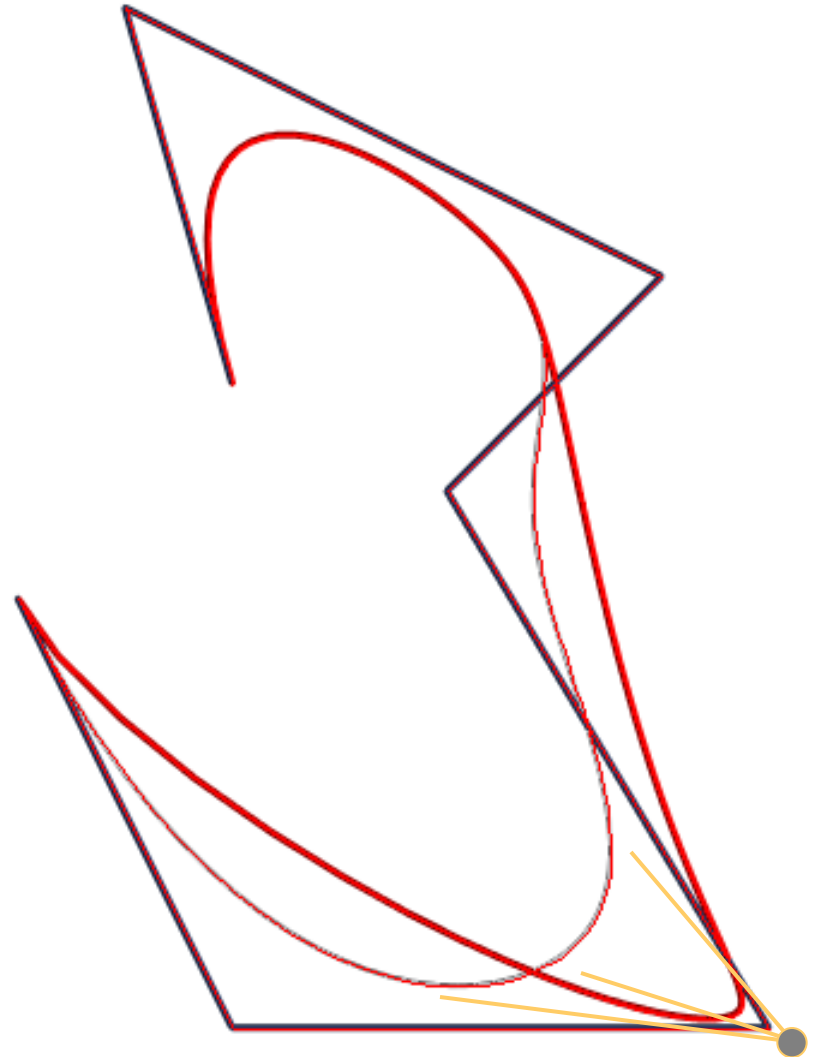


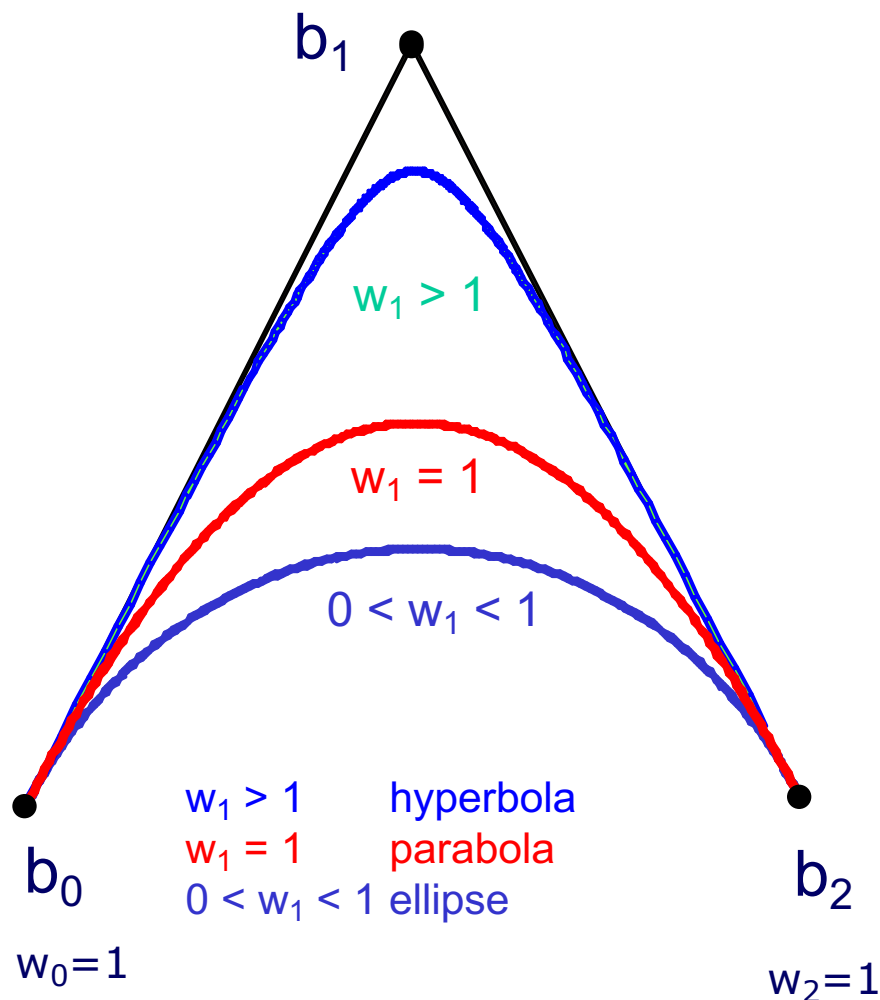


NURBS (non-uniform rational B-splines) have additional design parameters $\Rightarrow$ *weights*

Initially all weights equal to 1, and there the curve is an ordinary B-spline curve.

Increasing a weight of a control point pulls the curve towards that point; however, multiplication of all weights with the same factor has no effect.





Among the conics (circle, ellipse, parabola, hyperbola) only the parabola can be represented as a Bezier or a B-spline curve (degree 2). Using weights, all conics possess an exact NURBS representation



NURBS: NON UNIFORM RATIONAL B-SPLINE

A NURBS curve is defined as:

$$\mathbf{P}(t) = \frac{\sum_{i=0}^n N_{i,k}(t) w_i \mathbf{P}_i}{\sum_{i=0}^n N_{i,k}(t) w_i} \quad 0 \leq t \leq 1$$

- $\mathbf{P}_i$: control point;
- $\{\mathbf{P}_i: i = 0, 1, \dots, n\}$: control polygon;
- $\{w_i: i = 0, 1, \dots, n\}$: weights, $w_i > 0$ for each i ;
- $N_{i,k}(t)$ basis functions of the curve, given a vector of $m + 1$ non-periodic and non-uniform knots:

$$t_0 = t_1 = \dots = tk, t_{k+1}, \dots, t_{m-(k+1)}, t_{m-k} = \dots = t_{m-1} = t_m$$



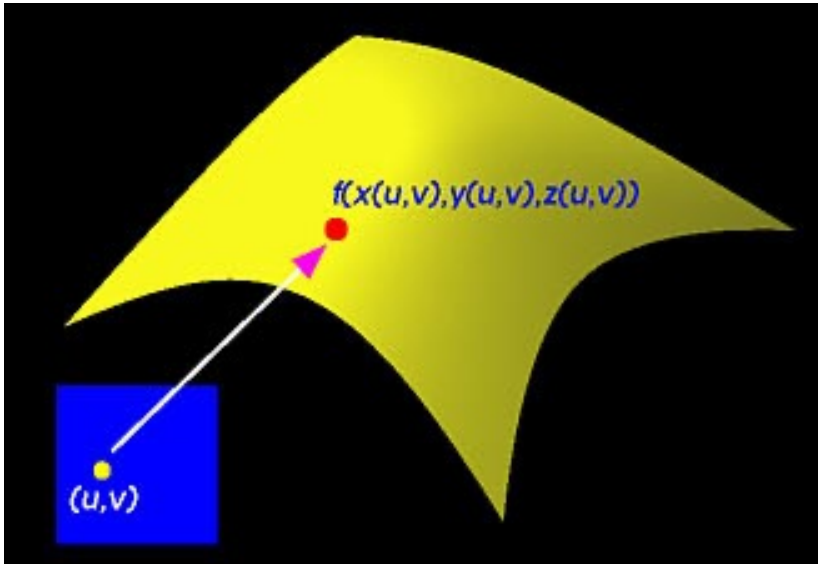
The functions:

$$R_{i,k}(t) = \frac{N_{i,k}(t)w_i}{\sum_{j=0}^n N_{j,k}(t)w_j}$$

Are called the rational basis functions of the curve:

$$\mathbf{P}(t) = \sum_{i=0}^n R_{i,k}(t)\mathbf{P}_i$$

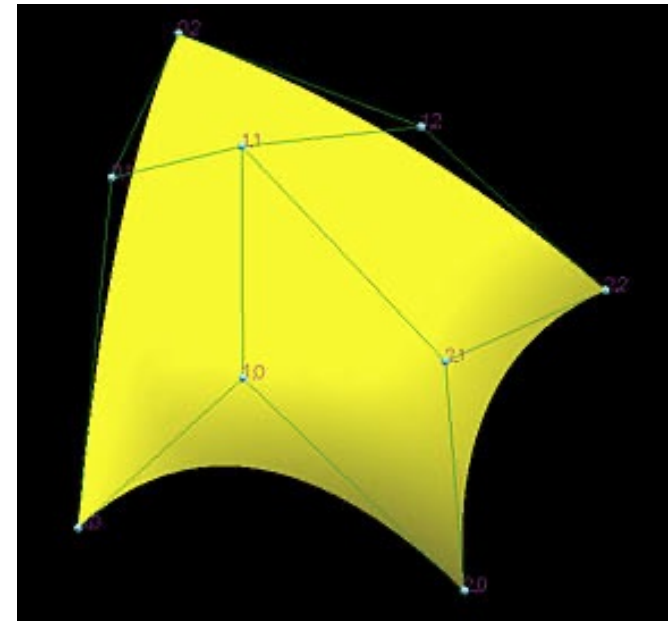
Parametric Surface



Bezier Surface

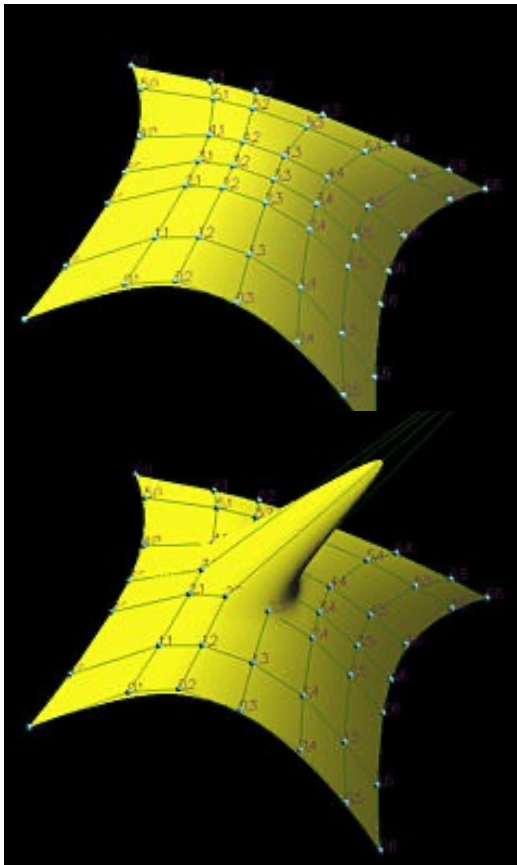
$$S(u, v) = \sum_{i=0}^n \sum_{j=0}^m B_{n,i}(u) B_{m,j}(v) \mathbf{P}_{i,j}$$

$$0 \leq u, v \leq 1$$



B-Spline

$$\mathbf{S}(u, v) = \sum_{i=0}^n \sum_{j=0}^m N_{i,k}(u) N_{j,q}(v) \mathbf{P}_{i,j}$$



NURBS

$$\mathbf{S}(u, v) = \sum_{i=0}^n \sum_{j=0}^m R_{i,j}(u, v) \mathbf{P}_{i,j}$$

$$R_{i,j}(u, v) = \frac{N_{i,k}(u) N_{j,q}(v) w_{i,j}}{\sum_{i=0}^n \sum_{j=0}^m N_{i,k}(u) N_{j,q}(v) w_{i,j}}$$

